PHYSICS

1) The position of a particle moving along a straight line is given by $x(t) = \frac{A}{B}(1-e^{-At})$ where A and B are constants. The dimensions of $\frac{A^3}{B}$ is same as

(1) linear momentum
(2) moment of inertia
(3) relative velocity
(4) acceleration

2) If a car covers $2/5^{th}$ of the total distance with v_1 speed and $3/5^{th}$ distance with v_2 speed, then average speed is :-

(1) $\frac{1}{2}\sqrt{v_1 v_2}$
(2) $\frac{5v_1 v_2}{3v_1 + 2v_2}$
(3) $\frac{2v_1 v_2}{v_1 + v_2}$
(4) $\frac{v_1 + v_2}{2}$

3) If the velocity of a particle is $(10 + 2t^2)$ m/s, then the average acceleration of the particle between 2s and 5s is :-

(1) 2 m/s^2
(2) 4 m/s^2
(3) 12 m/s^2
(4) 14 m/s^2

4) If the the relation between distance x and time t is of the form; $t = \alpha x^2 + \beta x$ α and β being appropriate constants, then the retardation of the particle is :-

(1) $2\alpha v^3$
(2) $2\beta v^3$
(3) $2\alpha\beta v^3$
(4) $2\beta^2 v^3$

5) Two identical charged ball of same mass and charge, each of 'q' are suspended from a common point by equal strings, each of length '□'. In equilibrium distance between the balls is 'a' then tension in string is :-

(1) $\frac{q^2\ell}{4\pi\in_0 a^3}$

(2) $\frac{q^2\ell}{2\pi\in_0 a^3}$

(3) $\frac{q^2\ell}{\pi\in_0 a^2}$

(4) $\frac{q^2\ell}{4\pi\in_0 a^2}$

6) When identical point charges are placed at the vertices of a cube of edge length a, each of them experiences a net force of magnitude F. Now these charges are placed on the vertices of another cube of edge length b. What will be magnitude of the net force on any of the charges ?
(These cubes are simply geometrical constructs and not made of any matter.)

(1) $\frac{a^2F}{b^2}$

(2) $\frac{b^2F}{a^2}$

(3) $\frac{aF}{b}$

(4) Insufficient information

7) **Assertion (A) :-** Two similarly charged bodies may attract each other. **Reason (R) :-** When charge on one body (Q) is much greater than that on another (q) and they are close enough to each other, then force of attraction between Q and induced charges exceeds the force of repulsion between Q and q.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

8)

If position of a particle is given as $x = 18t^2 - 2t^3 + 6t + 5$ then maximum velocity of particle is

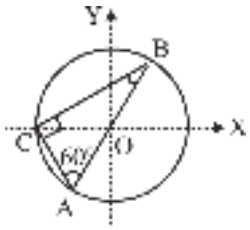
(1) 54

(2) 48

(3) 60

(4) 108

9) Consider a system of three charges $\frac{q}{3}, \frac{q}{3}$ and $-\frac{2q}{3}$ placed at points A, B and C respectively , as shown in the figure. Take O to be the centre of the circle of radius R and angle CAB = 60°. :-



(1) The electric field at point O is $\frac{q}{8\pi\varepsilon_0 R^2}$ directed along the negative x-axis.

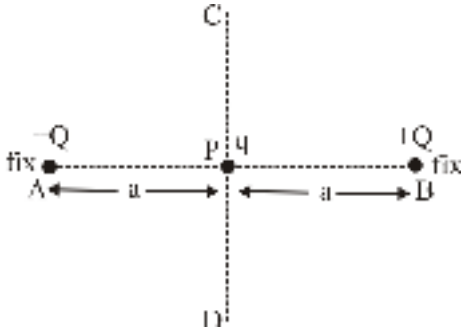
(2) The potential energy of the system is zero.

(3) The magnitude of the force between the charges at C and B is $\frac{q^2}{54\pi\varepsilon_0 R^2}$

(4) The potential at point O is $\frac{q}{12\pi\varepsilon_0 R}$

10)

For the arrangement shown below, choose the correct statement :-



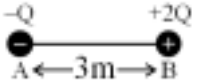
(1) If q is + ve, P is a position of stable equilibrium for displacement along AB

(2) If q is + ve, P is a position of unstable equilibrium for displacement along CD

(3) If q is – ve, P is a position of unstable equilibrium for displacement along AB

(4) All of the above

11) Charge 2Q and –Q are placed as shown in Fig. The point at which electric field intensity is zero will be :



(1) between –Q and 2Q

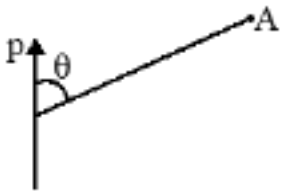
(2) on the left of –Q at distance $\left(3\sqrt{2}+3\right)$.

(3) on the left of 2Q at distance $\left(3\sqrt{2}+6\right)$.

(4) Both (2) & (3) are correct

12)

The electric field at *A* due to dipole *p* is perpendicular to $\vec{p}$. The angle θ is :-



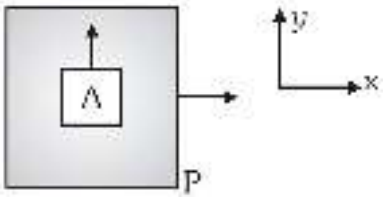
(1) 0°

(2) 90°

(3) $\tan^{-1}2$

(4) $\frac{\tan^{-1}}{\sqrt{2}}$

13) A block is moving along y–axis with velocity $\vec{v}_A = 4\hat{j}$ on a plank relative to ground and the plank is moving along x–axis with velocity $\vec{v}_P = 3\hat{i}$ at any instant of time. The unit vector in direction of friction force acting on block at this moment of time is :



(1) $\frac{3\hat{i}+4\hat{j}}{5}$

(2) $\frac{3\hat{i}-4\hat{j}}{5}$

(3) $\frac{-3\hat{i}+4\hat{j}}{5}$

(4) $\frac{-3\hat{i}-4\hat{j}}{5}$

14) If vector $\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$, then direction of cosine of the vector $\vec{A}$ are :

(1) $\frac{2}{\sqrt{45}}, \frac{4}{\sqrt{45}} \text{and} \frac{-5}{\sqrt{45}}$

(2) $\frac{1}{\sqrt{45}}, \frac{2}{\sqrt{45}} \text{and} \frac{3}{\sqrt{45}}$

(3) $\frac{4}{\sqrt{45}}, 0 \text{ and} \frac{4}{\sqrt{45}}$

(4) $\frac{3}{\sqrt{45}}, \frac{2}{\sqrt{45}} \text{and} \frac{5}{\sqrt{45}}$

15) If $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = -\hat{i} + 3\hat{j} + 4\hat{k}$, then projection of $\vec{A}$ on $\vec{B}$ will be :

(1) $\frac{3}{\sqrt{13}}$

(2) $\frac{3}{\sqrt{26}}$

(3) $\sqrt{\frac{3}{26}}$

(4) $\sqrt{\frac{3}{13}}$

16) If $\hat{A} = a\hat{i} + 0.8\hat{j}$ then find the value of a

(1) 0.6

(2) 0.06

(3) -0.6

(4) both (a) and (c)

17) Charges Q_1 and Q_2 lie inside and outside a closed surface S respectively. Let E be the field at any point on S and ϕ be the flux of E over S. Which statement is wrong?

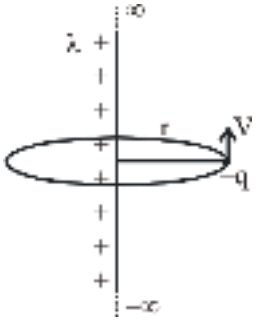
(1) If Q_1 changes, both E and ϕ will change

(2) If changes Q_2, E will change but ϕ will not change.

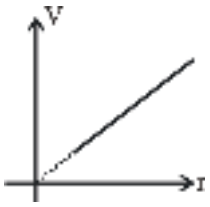
(3) If $Q_1 = 0$ and $Q_2 \neq 0$ then $E \neq 0$ but $\phi = 0$

(4) If $Q_1 = 0$ and $Q_2 = 0$ then $E = 0$ but $\phi \neq 0$

18) A charge – q of mass m is rotating in a circle of radius r around an infinitely large uniformly charged wire as shown. Speed of charge is V:-



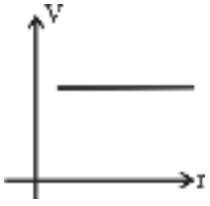
(1)



(2)



(3)



(4) 

19) $\int \left(\frac{\cos^2 x - \sin^2 x}{\sec^2 x - \tan^2 x} \right) dx$ is :-

(1) sin 2x + C

(2) 2 sin 2x + C

(3) $\frac{\sin 2x}{2} + C$

(4) $-\frac{\sin 2x}{2} + C$

20)

Use the approximation $(1+x)^n \approx 1 + nx, |x| << 1$, to find approximate value for $(999)^{1/3}$.

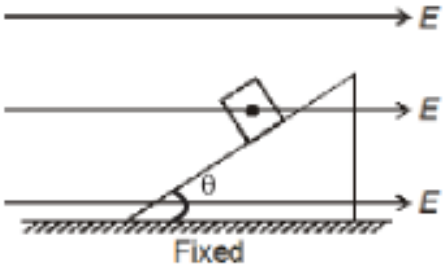
(1) $10 - \frac{1}{300}$

(2) $10 - \frac{1}{500}$

(3) $10 - \frac{1}{3000}$

(4) $10 - \frac{1}{30}$

21) A block of mass m (= 10 kg) and charge Q (= 2 C) is kept on smooth incline plane as shown. It is in equilibrium when a horizontal electric field $E = 20\frac{V}{m}$ is switched on in the region, then the angle of inclination of the incline plane is



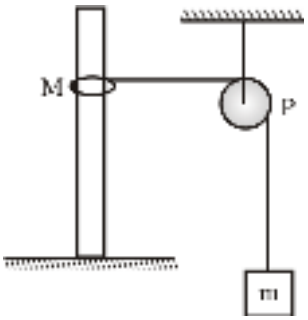
(1) $\theta = \tan^{-1}\left(\frac{5}{2}\right)$

(2) $\theta = \tan^{-1}\left(\frac{2}{5}\right)$

(3) $\theta = \cos^{-1}\left(\frac{2}{5}\right)$

(4) $\theta = \sin^{-1}\left(\frac{2}{5}\right)$

22) In the figure shown a ring of mass M and a block of mass m are in equilibrium. The string is light and pulley P does not offer any friction and coefficient of friction between pole and M is μ. The



frictional force acting on M is : -

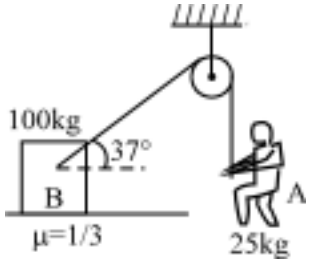
(1) Mg directed up

(2) μmg directed up

(3) (M –m)g directed down

(4) μmg directed down

23) Block B of mass 100 kg rests on a rough surface of friction coefficient μ = 1/3. A rope is tied to block B as shown in figure. The maximum acceleration with which boy A of 25 kg can climbs on



rope without making block move is :

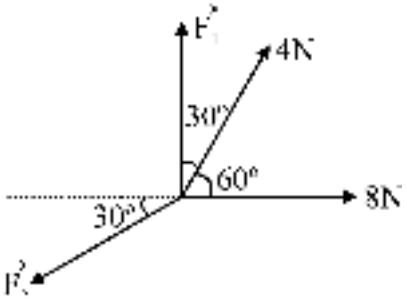
(1) $\frac{4g}{3}$

(2) $\frac{g}{3}$

(3) $\frac{g}{2}$

(4) $\frac{3g}{4}$

24) An object is in equilibrium under four concurrent forces in the directions as shown in the figure. Magnitudes of $\vec{F}_1$ and $\vec{F}_2$ are respectively. (Given : all the forces are in same plane) : -



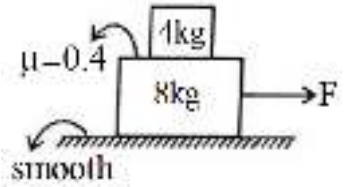
(1) $\frac{4}{\sqrt{3}}N$, $\frac{20}{\sqrt{3}}N$

(2) $\frac{20}{\sqrt{3}}N$, $\frac{4}{\sqrt{3}}N$

(3) 8N, 4N

(4) 4N, 8N

25) Find the maximum force for which both the blocks will move with same acceleration.



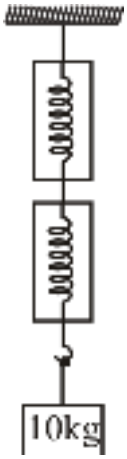
(1) 16 N

(2) 32 N

(3) 48 N

(4) 8 N

26) A block of mass 10 kg is suspended through two light spring balance as shown in figure :



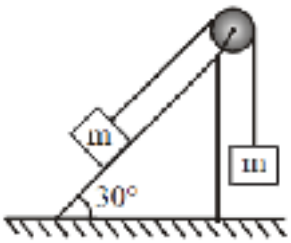
(1) Both the scales reads 10 kg–wt

(2) Both the scales reads 5 kg–wt

(3) The upper scale will read 10 kg–wt and the lower zero

(4) The reading may be any thing but their sum will be 10 kg–wt.

27) Two identical blocks each of mass m are shown in figure. Force exerted by clamp on the pulley is



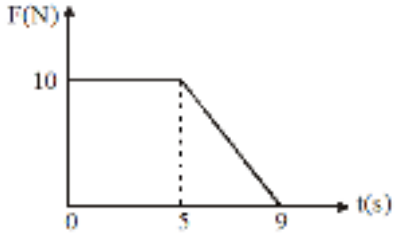
(1) $\frac{3\sqrt{3}mg}{4}$

(2) $\frac{3mg}{4}$

(3) $\sqrt{3}mg$

(4) $\frac{mg}{4}$

28) A body of mass 4 kg is acted on by a force which varies as shown in the graph below. The change in momentum is : -



(1) 280 N–s

(2) 140 N–s

(3) 70 N–s

(4) 210 N–s

29) Rectilinear motion of particle is defined by equation $v^3 = x^2$ (v = velocity). The acceleration of the particle when x = 8m is -

(1) $\frac{2}{3}m/s^2$

(2) $\frac{3}{4}m/s^2$

(3) $\frac{4}{3}m/s^2$

(4) None of these

30) A dipole p is placed along x-axis in electric field $E = \alpha x\hat{i}$. The net force on the dipole is:

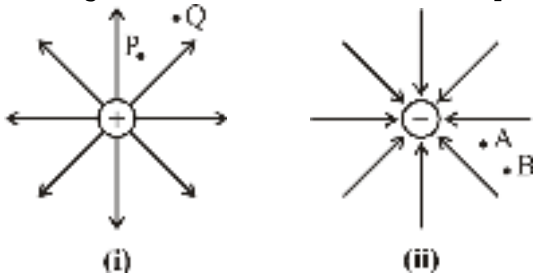
(1) $p\alpha^2\hat{i}$

(2) $p\alpha\hat{i}$

(3) $p\alpha\hat{j}$

(4) Zero

31) Figure shows the field lines of a positive and negative point charge respectively :



What is the sign of $U_P - U_Q$ and $U_B - U_A$ of a small negative charge :

(1) Negative, Negative

(2) Positive, Negative

(3) Negative, Positive

(4) Positive, Positive

32) A technician uses an analytical balance to measure three components of a chemical mixture. The recorded masses are 12.340 g, 0.56 g, and 105.2 g. Following the proper rules of experimental data reporting, what is the combined mass of this mixture expressed to the correct number of significant figures?

(1) 118.1 g

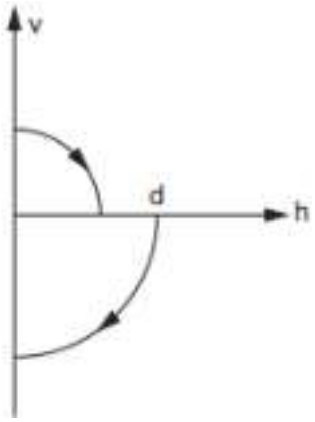
(2) 118.100 g

(3) 118.10 g

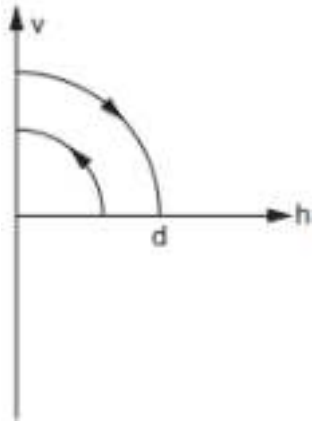
(4) 118 g

33) A ball is dropped vertically from height d above the ground. It hits the ground and bounces up vertically to a height 2d/3. Neglecting subsequent motion and air resistance, its velocity v varies with height h above the ground as :

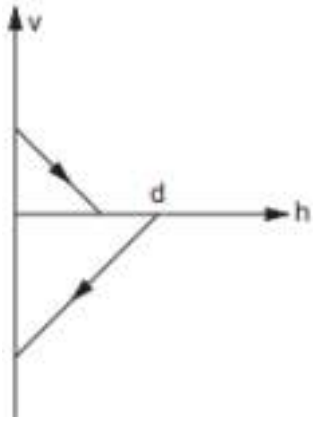
(1)



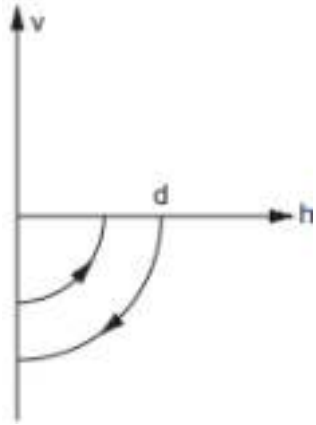
(2)



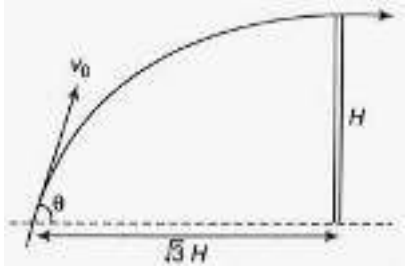
(3)



(4)



34) A projectile is thrown at an angle θ such that it is just able to cross a vertical wall at its highest point as shown in the figure. The angle θ at which the projectile is thrown is given by :



(1) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(2) $\tan^{-1}\sqrt{3}$

(3) $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(4) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

35) Trajectory of particle in a projectile motion is given as $y = x - \frac{x^2}{80}$. Here, x and y are in metres. For this projectile motion march the following with g = 10 m/s^2.

	Table-1		Table-2
(i)	Angle of projection	(p)	20 m
(ii)	Angle of velocity with horizontal after 4s	(q)	80 m

(iii)	Maximum height	(r)	45°
(iv)	Horizontal range	(s)	$\tan^{-1}\left(\frac{1}{2}\right)$

(1) (i)—(r), (ii)—(s), (iii)—(p), (iv)—(q)

(2) (i)—(r), (ii)—(r), (iii)—(p), (iv)—(q)

(3) (i)—(r), (ii)—(r), (iii)—(q), (iv)—(p)

(4) (i)—(r), (ii)—(s), (iii)—(q), (iv)—(p)

36)

A boat is sailing with a velocity $(3\hat{i} + 4\hat{j})$ with respect to ground and water in river is flowing with a velocity $(-3\hat{i} - 4\hat{j})$. Magnitude of relative velocity of the boat with respect to water is

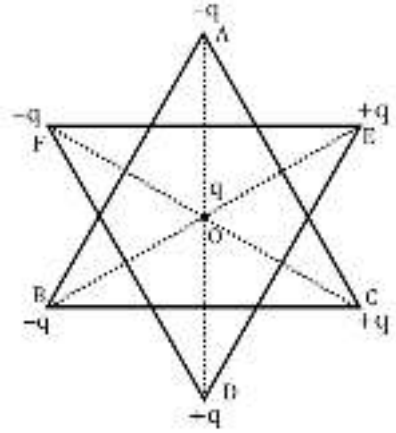
(1) 10

(2) 0

(3) 1

(4) 100

37) For given figure, resultant force on charge q situated at O will be (side length of both equilateral triangles ΔABC and ΔDEF is a)



(1) $\frac{4kq^2}{a^2}$

(2) $\frac{kq^2}{a^2}$

(3) $\frac{12kq^2}{a^2}$

(4) Zero

38) Three concentric spherical shells A, B and C of radii a, b and c (a < b < c) have surface charge densities σ, –σ and σ respectively. If V_A, V_B and V_C denote the potentials of the three shells, then, for c = a+b, we have :-

(1) $V_C = V_B = V_A$

(2) $V_C = V_A \neq V_B$

(3) $V_C = V_B \neq V_A$

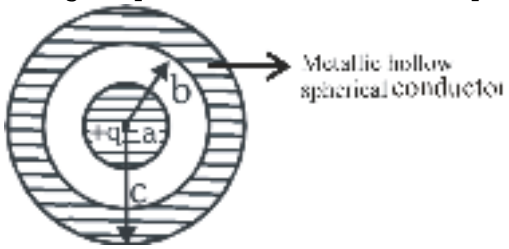
(4) $V_C \neq V_B \neq V_A$

39)

Electric potential at a point is: $\phi = x^2y + yz$. The electric field at the point (1, 3, 1) is

(1) 7 units
(2) 70 units
(3) 49 units
(4) 490 units

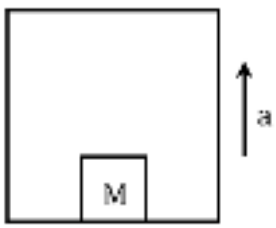
40) A solid non conducting sphere of radius 'a' has a net charge '+q' uniformly distributed throughout its volume. A hollow spherical conductor having inner and outer, radii 'b' and 'c' and net charge '–q' is concentric with the sphere (see the figure) then : -



(i) The electric field at a distance r from the centre of the sphere = $\frac{1}{4\pi\varepsilon_0}\frac{qr}{a^3}$ for r < a.
(ii) The electric field at distance r from the centre of the sphere for a < r < b = 0
(iii) The electric field at distance r from the centre of the sphere for b < r < c = 0
(iv) The charge on the inner surface of the hollow sphere = –q
(v) The charge on the outer surface of the hollow sphere = +q
Choose the correct options from below.

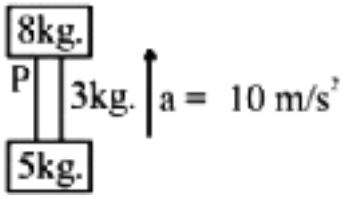
(1) (i) , (ii) and (v)
(2) (i) , (iii) and (iv)
(3) (ii) , (iii) and (iv)
(4) (ii) , (iii) and (v)

41) With what acceleration 'a' should the box of figure moving up so that the block of mass M exerts a force 7 Mg/4 on the floor of the box?



(1) g/4
(2) g/2
(3) 3g/4
(4) 4g

42) Two blocks of mass 8 kg. and 5 kg. are connected by a heavy rope of mass 3 kg. Complete system is accelerated upwards by 10 m/s^2 as shown in the figure. The tension at the point 'P' will be



:

(1) 60 N

(2) 90 N

(3) 160 N

(4) 150 N

43)

A physical quantity P = $\frac{\sqrt{abc^2}}{d^3e^{1/3}}$ is determined by measuring a, b, c, d and e separately with the percentage error of 2%, 3%, 2%, 1% and 6% respectively. Minimum amount of error is contributed by the measurement of :

(1) b

(2) a

(3) d

(4) e

44) The frequency of vibration f of a mass m suspended from a spring of spring constant K is given by a relation of this type, $f = Cm^xK^y$. Where C is a dimensionless quantity. The values of x and y are :-

(1) $x = \frac{1}{2}, y = \frac{1}{2}$

(2) $x = -\frac{1}{2}, y = -\frac{1}{2}$

(3) $x = \frac{1}{2}, y = -\frac{1}{2}$

(4) $x = -\frac{1}{2}, y = \frac{1}{2}$

45)

Velocity (in m/s) of a particle moving in a straight line is given by $V = (t^2 - 2t + 1)$:-

List-I		List-II	
(P)	Velocity (in m/s) of particle at t = 3 sec is	(1)	1
(Q)	Acceleration (in m/s^2) of particle at t = 2 sec is	(2)	2
(R)	Time (in sec) when particle is at rest is	(3)	3

(S)	Magnitude of average acceleration (in m/s^2) of particle in first one second is	(4)	4

(1) P→3, Q→2, R→4, S→1

(2) P→4, Q→3, R→2, S→1

(3) P→4, Q→2, R→1, S→1

(4) P→1, Q→4, R→3, S→2

CHEMISTRY

1) The rate of a reaction is expressed in different ways as follows :

$$+\frac{1}{2}\frac{d[C]}{dt} = -\frac{1}{3}\frac{d[D]}{dt} = +\frac{1}{4}\frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

The reaction is :

(1) 4A + B → 2C + 3D

(2) B + 3D → 4A + 2C

(3) A + B → C + D

(4) B + D → A + C

2) A hypothetical reaction, $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given ahead.

$A_2 \rightleftharpoons A + A$ .....(fast)

$A + B_2 \rightarrow AB + B$ .....(slow)

$A + B \rightarrow AB$ .....(fast)

The order of the reaction is :

(1) 2

(2) 1

(3) $\frac{3}{2}$

(4) Zero

3) The unit of rate constant obeying the rate expression $r = k[A]^1[B]^{2/3}$ is :

(1) $mol^{-2/3}$ $litre^{2/3}$ $time^{-1}$

(2) $mol^{2/3}$ $litre^{-2/3}$ $time^{-1}$

(3) $mol^{-5/3}$ $litre^{5/3}$ $time^{-1}$

(4) none of these

4) For a first order reaction, if the time taken for completion of 50% of the reaction is t second, the time required for completion of 99.9% of the reaction is :

(1) 5 t

(2) 10 t

(3) 100 t

(4) 2 t

5) For the first order reaction with rate constant k, which expression gives the half-life period (Initial concentration = a) :

(1) $\frac{\ell n 2}{k}$

(2) $\frac{1}{ka}$

(3) $\frac{0.693}{2k}$

(4) $\frac{3}{2ka^2}$

6) Half-lives of first order and zeroth order reactions are same. Ratio of rates at the start of reaction is :

(1) 0.693

(2) 2 × 0.693

(3) $\frac{1}{0.693}$

(4) $\frac{2}{0.693}$

7) The rate constant is numerically the same for three reaction of first, second and third order respectively. Which one is true for the three of reaction, if concentration of reactant is greater than 1M?

(1) $r_1 > r_2 > r_3$

(2) $r_1 = r_2 = r_3$

(3) $r_1 < r_2 < r_3$

(4) All of these

8) An elementary reaction between A and B is a second order reaction. Which of the following rate equations must be correct ?

(1) $r = k[A]^2 [B]^1$

(2) $r = k[A]^{3/2} [B]^{1/2}$

(3) $r = k[A]^2 [B]^2$

(4) $r = k[A][B]$

9) If the rate of gaseous reactions is independent of pressure, the order of reaction is :

(1) 0

(2) 1

(3) 2

(4) 3

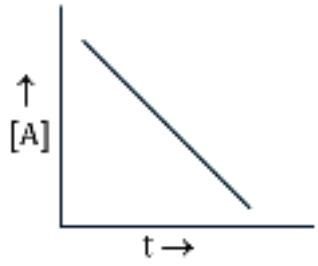
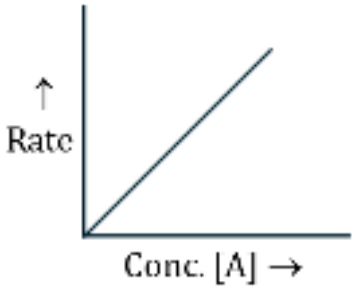
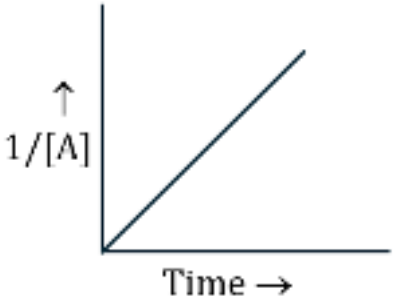
10) The rate constant of a reaction becomes equal to the pre-exponential factor when :

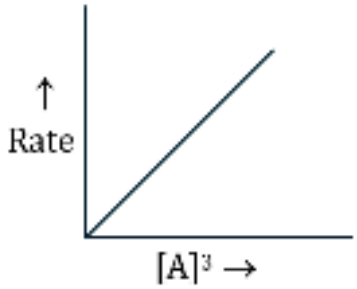
(1) the absolute temperature is infinity
(2) the temperature in celsius is zero
(3) the absolute temperature is zero
(4) the activation energy is infinity

11) Rate constant depends on :

(1) Temperature
(2) Concentration
(3) Pressure
(4) All of these

12) **Match the column-I with Column-II :**

Column I (Head)	**Column II (Order)**
(A) 	(p) 1
(B) 	(q) 2
(C) 	(r) 3

(D) 	(s) 0

(1) (A) → (p); (B) → (s); (C) → (r); (D) → (q)

(2) (A) → (s); (B) → (p); (C) → (q); (D) → (r)

(3) (A) → (p); (B) → (q); (C) → (r); (D) → (s)

(4) (A) → (r); (B) → (q); (C) → (p); (D) → (q)

13) Match the half-lives in Column-I with the orders in Column-II :

Column I (Half-life)	**Column II (Order)**
(A) $t_{1/2}$ = constant	(p) First order
(B) $t_{1/2} \propto a$	(q) Second order
(C) $t_{1/2} \propto \frac{1}{a}$	(r) Zero order

where, a = Initial concentration of the reactant

(1) A → p; B → r; C → q

(2) A → q; B → p; C → r

(3) A → r; B → q; C → p

(4) A → p; B → q; C → r

14) The reaction of A_2 and B_2 follows the equation

$A_2(g) + B_2(g) \rightarrow 2AB(g)$

The following data were observed

	$[A_2]_0$	**$[B_2]_0$**	**Initial rate of formation AB(g) (in Ms^{-1})**
(1)	0.10	0.10	2.5×10^{-4}
(2)	0.20	0.10	5×10^{-4}
(3)	0.20	0.20	10×10^{-4}

The value of order of the reaction is :

(1) 2

(2) 3

(3) 4

(4) 0

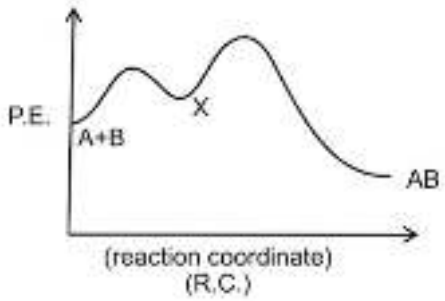
15) For an exothermic chemical process occurring in two steps as follows

(i) A + B → X (slow)

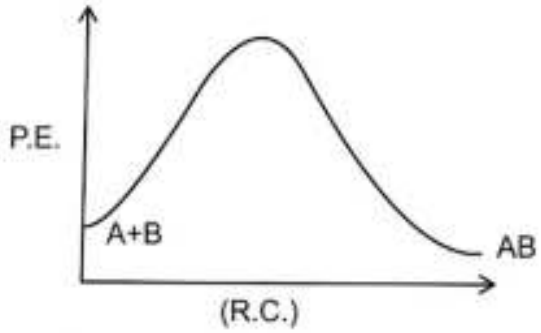
(ii) X → AB (fast)

The process of reaction can be best described by :

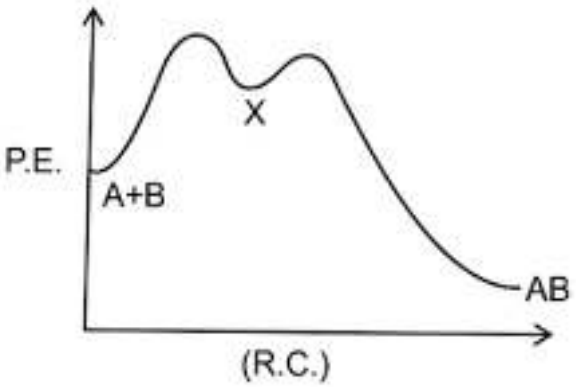
(1)



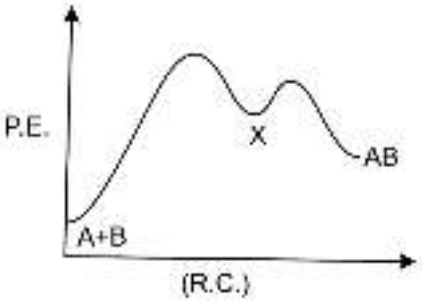
(2)



(3)



(4)



16) Match the Column-I with Column-II :

Column-I (Solution)	**Column-II (van't Hoff factor assuming complete dissociation/association)**
(A) $Ca_3(PO_4)_2$(aq.)	(p) i > 1
(B) $Al_2(SO_4)_3$(aq.)	(q) i = 5
(C) Urea	(r) i = 1
(D) CH_3COOH in benzene	(s) i < 1

(1) (A) → (p, s); (B) → (r); (C) → (q, r); (D) → (p)

(2) (A) → (p, q); (B) → (p, q); (C) → (r); (D) → (s)

(3) (A) → (p); (B) → (q); (C) → (s); (D) → (r)

(4) (A) → (s); (B) → (q); (C) → (r); (D) → (p)

17) Freezing point of an aqueous solution is –0.186°C.
Elevation of boiling point of the same solution is ..... if K_b = 0.512 K molality^{-1} and K_f = 1.86 K molality^{-1}:

(1) 0.186°C

(2) 0.0512°C

(3) 0.092°C

(4) 0.237°C

18) The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is:

(1) 0

(2) 1

(3) 2

(4) 3

19) A 5% (w/v) solution of cane sugar (molecular weight = 342) is isotonic with 1% (w/v) solution of a non-electrolyte substance X. The molecular weight of X is:

(1) 34.2

(2) 68.4

(3) 136.8

(4) 171.2

20) 0.5 M aqueous solution of glucose is isotonic with:

(1) 0.5 M KCl solution

(2) 0.5 M $CaCl_2$ solution

(3) 0.5 M urea solution

(4) 1 M solution of sucrose

21) Which of the following is dependent on temperature?

(1) Molarity

(2) Mole fraction

(3) Weight percentage

(4) Molality

22) Normality of 10% (w/v) H_2SO_4 solution nearly :

(1) 0.1

(2) 0.2

(3) 0.5

(4) 2

23) 5 N H_2SO_4 was diluted from 1 litre to 10 litres. Normality of the solution obtained is:

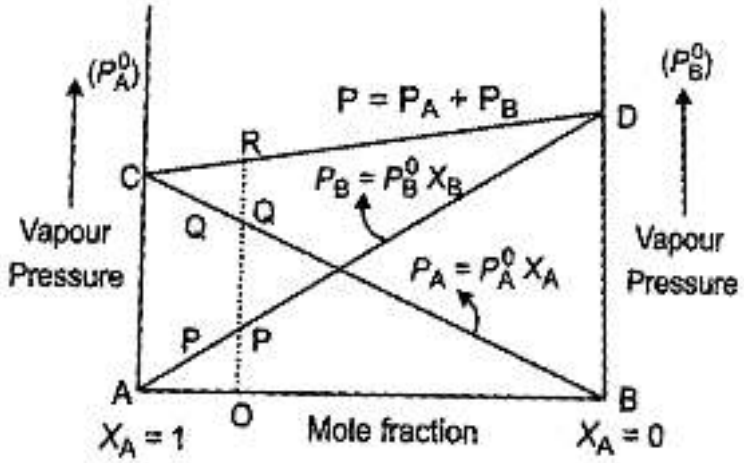
(1) 10 N

(2) 5 N

(3) 1 N

(4) 0.5 N

24) Two volatile liquids A and B form ideal solution.
Considering the following vapour-pressure composition graph.



OR will be equal to :

(1) OQ + QR

(2) OQ + PQ

(3) OP + OQ

(4) OP + PR

25) Which one is not equal to zero for an ideal solution:

(1) ΔS_{mix}

(2) ΔV_{mix}

(3) ΔH_{mix}

(4) All of these

26) For an ideal solution of two components A and B, which of the following is true?

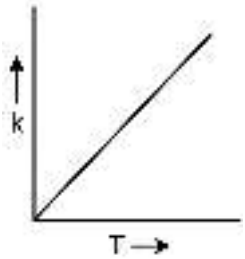
(1) $\Delta H_{mixing} > 0$ (zero).

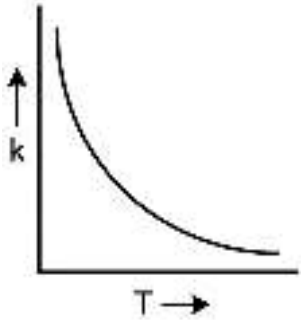
(2) A–B interaction is stronger than A–A and B–B interactions.

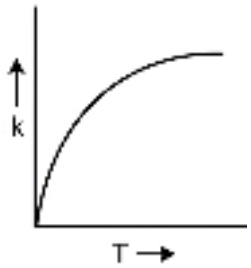
(3) A–A, B–B and A–B interactions are identical.

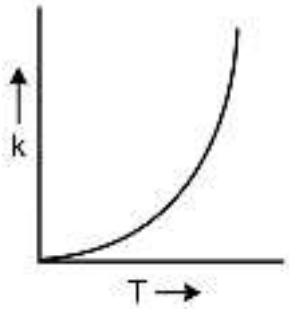
(4) $\Delta H_{mixing} < 0$ (zero)

27) The rate constant of most of the reactions increases with increase in temperature. According to the Arrhenius equation, $k = Ae^{-E_a/RT}$. The curve of rate constant 'k' against temperature 'T' will be :

(1) 

(2) 

(3) 

(4) 

28) The rate constant of a reaction is 2.1×10^{-2} mol^{-2} $litre^2$ min^{-1}. The order of reaction is:

(1) 1

(2) 2

(3) 3

(4) zero

29) The chemical reaction, $2O_3 \rightarrow 3O_2$ proceeds as follows:

$O_3 \rightleftharpoons O_2 + O$ (fast)

$O + O_3 \rightarrow 2O_2$ (slow)

The rate law expression should be:

(1) $r = k[O_3]^2$

(2) $r = k[O_3][O_2]$

(3) $r = [O_3][O_2]^2$

(4) $r = k[O_3]^2[O_2]^{-1}$

30) The temperature coefficient for most of the reactions lies between:

(1) 1 and 3

(2) 2 and 3

(3) 1 and 4

(4) 2 and 4

31) The reaction $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ is of first order. If volume of reaction vessel is reduced to $\frac{1}{3}$, the rate of reaction would be :

(1) $\frac{1}{3}$ times

(2) $\frac{2}{3}$ times

(3) 3 times

(4) 6 times

32) If in a 1st order reaction 75% of reaction complete in 4 hrs. Then how much time is required to complete 87.5% of reaction?

(1) 32 h

(2) 6 h

(3) 8 h

(4) 16 h

33) When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is :

(1) first

(2) second

(3) more than zero but less than first

(4) zero

34) Slope of the straight line obtained by plotting $\log_{10}$ k against 1/T represents which term?

(1) $-E_a$

(2) $-2.303\ E_a / R$

(3) $-E_a / (2.303\ R)$

(4) $-E_a / R$

35) If 18 g of glucose is present in 1000 g of solvent, the solution is said to be:

(1) 1 molar

(2) 0.1 molar

(3) 0.5

(4) 0.1 molal

36) The depression in f.pt. of 0.01 m aqueous solution of urea, sodium chloride and sodium sulphate is in the ratio:

(1) 1 : 1 : 1

(2) 1 : 2 : 4

(3) 1 : 2 : 3

(4) 2 : 2 : 3

37) Vapour pressure of a solution containing non-volatile solute is:

(1) inversely proportional to the mole fraction of the solute

(2) directly proportional to the mole fraction of the solvent

(3) inversely proportional to the mole fraction of the solvent

(4) directly proportional to the mole fraction of solute

38) Value of Henry's constant (K_H)__.

(1) decreases with increase in temperature

(2) increases with increase in temperature

(3) remains constant

(4) first increase then decreases

39) If molality of the dilute solutions is doubled, the value of molal depression constant (K_f) will be:

(1) halved

(2) tripled

(3) unchanged

(4) doubled

40) The mixture of two liquids boils without changing its composition is called:

(1) zeotropic liquid mixture

(2) binary liquid mixture

(3) azeotropic liquid mixture

(4) stable structure complex

41) What is the freezing point of a solution that contains 10.0 g of a glucose ($C_6H_{12}O_6$) in 100 g of H_2O ?
K_f = 1.86° C/m.

(1) –0.186º C

(2) +0.186º C
(3) –0.10º C
(4) –1.03º C

42) Which of the following is not a non-electrolyte?

(1) Acetic acid
(2) Glucose
(3) Ethanol
(4) Urea

43) Pure water can be obtained from sea water by:

(1) centrifugation
(2) plasmolysis
(3) reverse osmosis
(4) sedimentation

44) What is the van't Hoff factor for sodium phosphate?

(1) 1
(2) 2
(3) 3
(4) 4

45) Van't Hoff factor more than unity indicates that the solute in solution has:

(1) associated
(2) dissociated
(3) both
(4) cannot say anything

BIOLOGY

1) Match the following epithelial tissues with their locations/functions:

Column I (Epithelial tissue)	**Column II (Location / Function)**
A. Squamous epithelium	i. Intestine, stomach – secretion & absorption
B. Cuboidal epithelium	ii. Alveoli, blood vessels – diffusion
C. Columnar epithelium	iii. Kidney tubules, ducts – secretion & absorption
D. Ciliated epithelium	iv. Bronchioles, fallopian tube – movement of mucus/ovum

(1) A–ii, B–iii, C–i, D–iv
(2) A–iii, B–ii, C–iv, D–i

(3) A–i, B–iv, C–ii, D–iii

(4) A–iv, B–i, C–iii, D–ii

2) Arrange the following events in correct sequence regarding secretion in exocrine glands:
a. Secretion of product by glandular cells.
b. Passage through ducts or tubes.
c. Release into specific target site (like buccal cavity).
d. Storage in secretory vesicles inside cells.

(1) d → a → b → c

(2) a → d → b → c

(3) d → b → a → c

(4) b → a → d → c

3) State whether the following statements are Correct or Incorrect. .
1. Areolar tissue acts as a packing tissue beneath the skin and contains fibroblasts, macrophages, and mast cells.
2. Adipose tissue cells are specialized to store proteins in the form of reserve food.
3. Dense regular connective tissue has collagen fibres arranged in a parallel pattern, found in tendons and ligaments.
4. Dense irregular connective tissue has fibroblasts and many collagen fibres oriented in different directions, present in skin.
5. Connective tissues have no role in secretion of fibres or ground substance.

(1) 2, 3, and 4 are incorrect; rest are correct

(2) 1 and 4 are correct; rest are incorrect

(3) 2 and 5 are incorrect; rest are correct

(4) 2 and 5 are correct; rest are incorrect

4) Match the tissues with their features:

Tissues	Features
A. Cartilage	i. Hard, non-pliable ground substance rich in calcium salts and collagen fibres, provides structural support.
B. Bone	ii. Solid, pliable tissue with chondrocytes in cavities, resists compression, found in nose & joints.
C. Blood	iii. Fluid connective tissue with plasma, RBCs, WBCs, and platelets, transports substances.

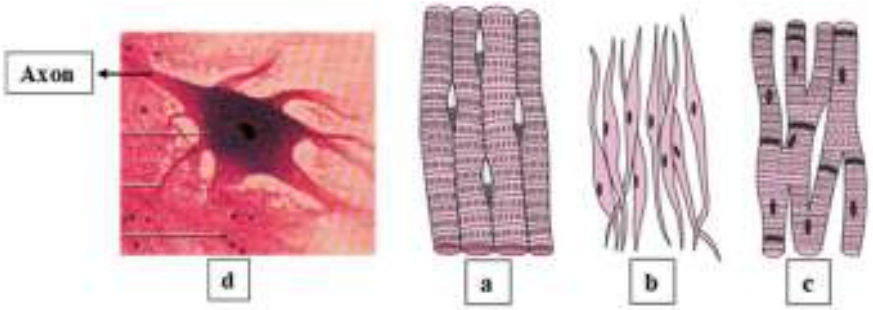
(1) A–ii, B–iii, C–i

(2) A–i, B–ii, C–iii

(3) A–iii, B–ii, C–i

(4) A–ii, B–i, C–iii

5) Refer to Figure a, b, c and d and answer the following:



1. The muscle fibre with striations and multiple nuclei is _____.
2. The muscle tissue showing junctions between adjacent cells is _____.
3. The fusiform-shaped fibres without striations belong to _____.
4. In the neuron, the part labeled “axon” is responsible for _____.

(1) a-1-Skeletal muscle; c-2-Cardiac muscle; b-3-Smooth muscle; d-4-Conducting impulses away from cell body

(2) a-1-Cardiac muscle; b-2-Skeletal muscle; c-3-Smooth muscle; d-4-Receiving impulses from dendrites

(3) a-1-Smooth muscle; b-2-Cardiac muscle; c-3-Skeletal muscle; d-4-Supporting neuron functions

(4) a-1-Skeletal muscle; b-2-Smooth muscle; c-3-Cardiac muscle; d-4-Carrying impulses toward the cell body

6) State whether the following are True (T) or False (F):
1. Glandular epithelium is always multicellular.
2. Tight junctions prevent leakage across tissues.
3. Endocrine glands release hormones directly into body fluids.
4. Cuboidal epithelium lines ducts of glands and parts of nephrons.
5. Compound epithelium is single-layered and mainly absorbs nutrients.

(1) T, T, F, T, T

(2) F, T, T, T, F

(3) F, T, T, F, F

(4) F, T, T, F, T

7) **Assertion (A):** Areolar connective tissue provides support framework for epithelium.
Reason (R): Areolar tissue contains fibroblasts, macrophages, and mast cells.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, R is false.

(4) A is false, R is true.

8) Which connective tissue is found in ligaments?

(1) Dense regular connective tissue

(2) Dense irregular connective tissue

(3) Areolar connective tissue

(4) Adipose connective tissue



(a)



(b)

9) Refer to Figure a and b in the diagram :
Which of the following statements are correct?
1. ‘a’ has fibroblasts that secrete collagen and elastin fibres.
2. ‘a’ is specialised to store fats in fat storage areas.
3. Adipocytes are absent in 'b'.
4. Macrophages are present in ‘a’ for immune defence.

(1) 1 and 4 only

(2) 1, 3, and 4 only

(3) 2 and 3 only

(4) All 1, 2, 3, and 4

10) Refer to the given morphology of cockroach and match correctly:

Column-I	Column-II
1. Filiform antennae	A. First pair of wings, leathery, covering hindwings
2. Compound eyes	B. Sensory perception of environment
3. Mesothoracic leg	C. Vision with many ommatidia
4. Tegmina	D. Locomotion

(1) 1–A, 2–D, 3–B, 4–C

(2) 1–C, 2–B, 3–A, 4–D

(3) 1–D, 2–A, 3–C, 4–B

(4) 1–B, 2–C, 3–D, 4–A

11) State whether the following are True (T) or False (F):
1. The cockroach head capsule bears a pair of compound eyes and ocelli.
2. Antennae arise from membranous sockets and act as sensory organs.
3. Labium forms the upper lip while labrum forms the lower lip.
4. Hypopharynx acts like a tongue and lies between maxillae and mandibles.
5. Cockroach mouthparts are of biting and chewing type.

(1) F T F F T

(2) T T F T T

(3) F F F T F

(4) T T F F T

12) Which of the following statements are correct?
1. The storage organ before the gizzard is called the crop.
2. Malpighian tubules help in digestion of food.
3. The circulatory system of cockroach is of closed type.

4. Haemolymph of cockroach is colourless and lacks respiratory pigments.
5. The heart pumps blood anteriorly through the aorta.

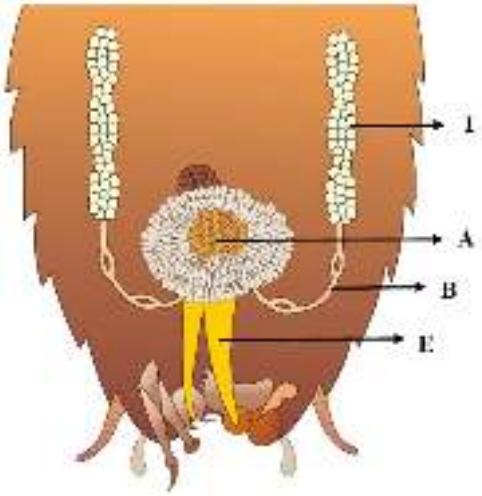
(1) 1, 2, 3 only

(2) 1, 3, 4, 5 only

(3) 1, 4, 5 only

(4) 2, 3, 5 only

13) Refer to the diagram of the reproductive system of cockroach.



Arrange the following structures labelled T, A, B, E in the correct sequence of sperm transport in male cockroach:

(1) T → B → A → E

(2) B → E → A → T

(3) T → A → B → E

(4) A → T → B → E

14) State whether the following statements are True (T) or False (F):
1. Cockroach head is formed by the fusion of 6 segments.
2. Antennae of cockroach arise from sockets located behind the eyes.
3. The first pair of wings arises from the mesothorax.
4. In females, the 7th sternum is boat-shaped, forming a brood pouch with the 8th and 9th sterna.
5. Males bear short thread-like anal styles which are absent in females.

(1) F, F, F, T, T

(2) T, T, F, T, F

(3) T, F, T, F, T

(4) T, F, T, T, T

15) Which of the following is the correct sequence of structures food passes through in cockroach?

(1) Mouth → Oesophagus → Crop → Gizzard → Hepatic caeca → Midgut → Rectum → Anus

(2) Mouth → Pharynx → Oesophagus → Crop → Gizzard → Midgut → Ileum → Colon → Rectum → Anus

(3) Mouth → Crop → Gizzard → Pharynx → Oesophagus → Midgut → Colon → Rectum

(4) Mouth → Pharynx → Crop → Oesophagus → Gizzard → Ileum → Anus

16) Fill in the blanks with suitable words from the frog digestive system:
(i) The liver secretes _____ which is stored in the _____.
(ii) The _____ produces pancreatic juice containing digestive enzymes.
(iii) Undigested food passes into the _____ and finally opens outside by the cloaca.

(1) Bile, Gall bladder; Pancreas; Rectum

(2) Bile, Pancreas; Liver; Cloacal aperture

(3) HCl, Gall bladder; Pancreas; Stomach

(4) Bile, Stomach; Gall bladder; Rectum

17) Consider the following statements about frog physiology:
1. During aestivation and hibernation, frogs respire mainly through lungs.
2. Frog's heart has three chambers: two atria and one ventricle.
3. The hepatic portal system connects the kidney and liver.
4. Frogs are ureotelic, i.e., they excrete nitrogenous waste mainly in the form of urea.
5. RBCs of frogs are nucleated and contain haemoglobin.
Which of the above statements are correct?

(1) 1, 2 and 5 only

(2) 2, 3, 4 and 5 only

(3) 1, 3 and 4 only

(4) 2, 4 and 5 only

18) Match the parts of the frog nervous system/brain in Column I with their correct combinations in Column II:

Column I (Part)	**Column II (Function)**
A. Forebrain	i. Controls balance and equilibrium
B. Midbrain	ii. Controls voluntary actions, intelligence, smell
C. Hindbrain (Cerebellum)	iii. Pair of optic nerves
D. Medulla oblongata	iv. Controls involuntary activities (heartbeat, respiration)

(1) A–ii, B–iii, C–i, D–iv

(2) A–iii, B–ii, C–iv, D–i

(3) A–ii, B–iv, C–iii, D–i

(4) A–iv, B–ii, C–iii, D–i

19) Identify the correct (C) and incorrect (I) statements regarding frog reproduction:
1. Male frogs have 10–12 vasa efferentia arising from testes which join Bidder's canal.
2. In female frogs, the oviducts open into cloaca separately without connection to kidneys.
3. Fertilization in frogs is internal and occurs inside the female cloaca.
4. A female frog can lay nearly 2500–3000 ova at a time.
5. External ear and tympanum both help in balancing (equilibrium).

(1) C → 1, 2, 4 ; I → 3, 5

(2) C → 2, 3, 4 ; I → 1, 5

(3) C → 1, 3, 4 ; I → 2, 5

(4) C → 1, 2, 3 ; I → 4, 5

20) **Assertion (A):** Hind limbs of frogs are longer and stronger than forelimbs.
Reason (R): Hind limbs help mainly in burrowing, while forelimbs are used for swimming.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

21) Which of the following cell junctions help to stop substances from leaking across a tissue?

(1) Gap junctions

(2) Tight junctions

(3) Adhering junctions

(4) Intercalated discs

22) Cuboidal epithelium is found in the :-

(1) Lining of stomach

(2) Fallopian tubes in female

(3) PCT of nephron in kidney

(4) Walls of blood vessels

23) Which type of tissue forms the inner lining of blood vessels?

(1) Cuboidal epithelium

(2) Columnar epithelium

(3) Squamous epithelium

(4) Ciliated epithelium

24) Match column I and column II

	Column I		**Column II**
a.	Anal cerci	(i)	4-6th abdominal segment
b.	Anal style	(ii)	6-7th abdominal segment
b.	Testis	(iii)	10th tergum
d.	Ovary	(iv)	9th Sternum
e.	Mushroom gland	(v)	2-6th abdominal segment

(1) a-iii, b-iv, c-ii, d-v, e-i

(2) a-iii, b-iv, c-i, d-v, e-ii

(3) a-iv, b-iii, c-i, d-v, e-ii

(4) a-iii, b-iv, c-v, d-i, e-ii

25) Regarding the cockroach, read the statements and mark how many of the following are correct?
I. The foregut is lined by cuticle and consists of the pharynx, oesophagus, crop and gizzard.
II. The haemolymph, a colorless fluid, circulates freely throughout the body cavity.
III. The malpighian tubules are excretory organs at the junction of midgut and hindgut
IV. A series of segmental pores, called sclerites, are present on the thorax and abdomen.

(1) One
(2) Two
(3) Three
(4) Four

26) Where are sclerites present in a cockroach?

(1) Only on the head
(2) On all body segments
(3) Only on the thorax
(4) On the abdomen and wings

27) The mouth part of a cockroach that is comparable to our tongue is:

(1) Mandible
(2) Labium
(3) Hypopharynx
(4) Maxillae

28) The process by which a tadpole develops into an adult frog is called:

(1) Metamorphosis
(2) Metamerism
(3) Regeneration
(4) Parthenogenesis

29) What is the difference between cutaneous and pulmonary respiration in a frog?

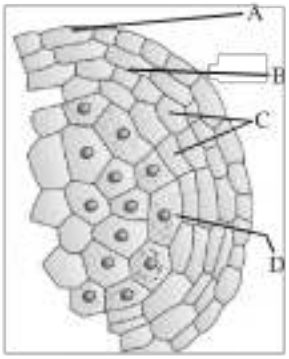
(1) Cutaneous respiration uses lungs, while pulmonary respiration uses skin.
(2) Cutaneous respiration occurs through the skin, while pulmonary respiration occurs through the lungs.
(3) Cutaneous respiration is for oxygen, while pulmonary respiration is for carbon dioxide.
(4) Cutaneous respiration is for land, while pulmonary respiration is for water.

30) Which portal system present in frog ?

(1) Both hepatic and renal portal systems
(2) Renal portal system
(3) Hepatic portal vein

(4) No portal system present

31) Given below is the diagrammatic sketch of a transverse section of young anther. Identify the



parts labelled A, B, C and D and select the right option about them.

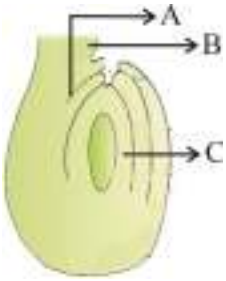
	Part-A	**Part-B**	**Part-C**	**Part-D**
(1)	Epidermis	Middle layers	Endothecium	Tapetum
(2)	Epidermis	Endothecium	Middle layers	Tapetum
(3)	Exodermis	Endothecium	Middle layers	Tapetum
(4)	Epidermis	Endodermis	Middle layers	Sporogenous tissue

(1) 1

(2) 2

(3) 3

(4) 4

32) 

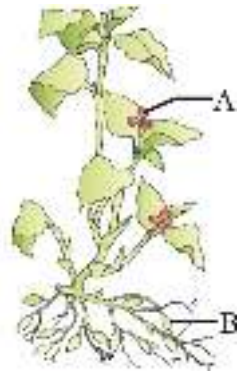
Select the correct match with respect to above figure :-

(1) A-Funicle, B-Hilum, C-Embryosac

(2) A-Hilum, B-Embryosac, C-Funicle

(3) A-Hilum, B-Funicle, C-Nucellus

(4) A-Nucellus, B-Hilum, C-Funicle

33) 

Select the correct statement with respect to above figure :-

(1) A, always forms seed even in absence of pollinator

(2) B, always need pollinator

(3) B, Prevent xenogamy but not autogamy

(4) A, may prevent both autogamy and xenogamy

34) Given below are two statements :
Statement-I :- A typical angiospermic anther is bilobed and tetrasporangiate.
Statement-II :- In a transverse section, a typical microsporongium of angiosperms appears near circular in outline.

(1) Both statement-I and II are incorrect.

(2) Statement-I is incorrect but statement-II is correct.

(3) Statement-I is correct but statement-II is incorrect.

(4) Both statement-I & II are correct.

35) Below two statement are given :
Statement-I :- The proximal end of the filament is attached to the thalamus or the petal of the flower.
Statement-II :- A typical angiosperm anther is bilobed with each lobe having two theca.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

36) Below two statements are given.
Statement-I :- In a normal flower, which opens and exposes the anthers and the stigma, complete autogamy is rather rare.
Statement-II :- Autogamy in flowers do not requires synchrony in pollen release and stigma receptivity.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

37) **Statement-I :-** Polar nuclei are situated below the egg apparatus in the large central cell

Statement-II :- In majority of flowering plants, one of the megaspores is functional while three degenerate.

(1) Both statements are correct
(2) Both statements are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

38) The pollen viability in some cereals like wheat and rice is with in :-

(1) 30 minutes
(2) 30 seconds
(3) 30 Hours
(4) 30 Days

39) Which of the following statements are true?
(i) Main plant body of angiosperms is a gametophyte
(ii) Thalamus is a modified stem
(iii) Angiosperm bears flowers and fruits.
(iv) Angiospermic endosperm is haploid

(1) i, iii only
(2) Only ii
(3) ii, iii, only
(4) i, ii, iv only

40) In flowering plants male gametes are formed by:-

(1) Generative cell
(2) Body cell
(3) Synergid cell
(4) Tube cell

41) In majority of angiosperm female gametophyte is :-

(1) 7 celled 7 nucleated structure
(2) 7 celled 8 nucleated structure
(3) 7 celled 6 nucleated structure
(4) 4 celled 4 nucleated structure

42) **Assertion :-** Cereals such as Rice and Wheat pollen grain lose viability within 30 minutes of their release.
Reason :- Pollen of Rosaceae, Leguminoseae and Solanaceae maintain viability for 30 minutes only.

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.

(4) A is false but R is true.

43) **Assertion:** Pollen tablets is used as a food supplement.
Reason: Pollen grains are rich in nutrients.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

44) The plants in which both autogamy and geitonogamy are prevented :

(1) Castor & Coconut

(2) Datepalm & Maize

(3) Papaya & Cucumber

(4) Datepalm & Papaya

45) There is a need for emasculation in Artificial hybridisation if the

(1) Female parent produces bisexual flowers

(2) Male parent produces bisexual flowers

(3) Female parent produces unisexual flowers

(4) Male parent produces unisexual flowers

46) Features seen in flowers which are pollinated by Insects are
a. Non-sticky pollen
b. Colourful flowers
c. Well exposed stamens
d. Secrete foul odour
e. Large flowers
f. Mucilaginous covering around pollen grain

(1) b, d and e

(2) a, b and d

(3) c, d and f

(4) a, c and f

47) Assured seed set even in absence of pollinators, occurs in

(1) Amorphophallus

(2) Ficus

(3) Commelina

(4) Vallisneria

48) Well exposed stamens, large feathery stigma, single ovule in each ovary and numerous flowers

packed into inflorescence is characteristic features of

(1) Entomophily
(2) Anemophily
(3) Hydrophily
(4) Zoophily

49) **Assertion :** Nectar and pollen grain are not usual floral rewards.
Reason : For harvesting the reward from flowers, animal visitors come in contact with anther and stigma.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

50) Sequence of development during the formation of embryosac is –

(1) Archesporium → megaspore → megaspore mother cell
(2) Megaspore mother cell → archesporium → megaspore → embryosac
(3) Megaspore → megaspore mother cell → archesporium → embryosac
(4) Archesporium → megaspore mother cell → megaspore → embryosac

51) Which of the following gives the correct categorisation of three plants according to the type of pollination (A, B, C) they give out.

	A Hydrophily	**B Anemophily**	**C Zoophily**
(1)	*Zostera*	*Amorphophallus*	Grasses
(2)	Grasses	*Amorphophallus*	*Zostera*
(3)	*Zostera*	Grasses	*Amorphophallus*
(4)	Amorphophallus	Grasses	*Zostera*

(1) 1
(2) 2
(3) 3
(4) 4

52) The two cells at which pollen grains are discharged are :-

(1) Larger generative cell and smaller vegetative cell
(2) Both large generative cell
(3) Both smaller vegetative cell
(4) Smaller generative cell and larger vegetative cell

53) The devices developed by plants to discourage self pollination include

(A) In some species the pollen release and receptive stigma are not synchronized
(B) In some species the anther and stigma are placed at different positions such that they do not come in contact
(C) Self - incompatibility

(1) A and B only

(2) A only

(3) A, B and C

(4) only B

54) Which of the following statements is incorrect for aquatic plants ?
(A) Vallisneria and hydrilla are freshwater plants pollinated by wind.
(B) Zostera a marine grass uses water as its pollinating agent.
(C) Water hyacinth and water Lily are pollinated by insects or wind.
(D) Most of the aquatic plant flowers emerges out of water and are pollinated by wind or insects.

(1) All are correct except A

(2) All are correct except B and C

(3) All are correct

(4) All are incorrect

55) Match column I with column II and select the correct option from codes given below.

Column I		**Column II**	
(A)	Cleistogamy	(i)	Insect pollination
(B)	Geitonogamy	(ii)	Bud pollination
(C)	Entomophily	(iii)	Pollination between flowers in the same plants
(D)	Xenogamy	(iv)	Wind pollination
		(v)	Cross pollination

(1) A - (iii), B - (i), C - (v), D - (ii)

(2) A - (i), B - (v), C - (ii), D - (iii)

(3) A - (ii), B - (iii), C - (i), D - (v)

(4) A - (v), B - (iv), C - (iii), D - (ii)

56) Match the pollination types with their correct examples:

	Column A (Pollination type)		**Column B (Example)**
A.	Entomophily	P.	Amorphophallus
B.	Anemophily	Q.	Maize
C.	Hydrophily (epihydrophily)	R.	Vallisneria
D.	Hydrophily (Hypohydrophily)	S.	Zostera

(1) A-P, B-Q, C-R, D-S

(2) A-R, B-P, C-S, D-Q

(3) A-S, B-Q, C-P, D-R

(4) A-Q, B-R, C-S, D-P

57)

In the list given below how many structures are haploid ?
Pollen mother cell, Antipodal cell, Male gamete, Pollen grain, Synergid, Nucellus, Funicle, Perisperm, Integuments, Megaspore, Microspore tetrad.

(1) Five

(2) Six

(3) Four

(4) Seven

58) Which of the following is not true regarding the flowers?

(1) Flowers have aesthetic, cultural, religious and social values,

(2) Flowers are unit of sexual reproduction

(3) Flowers have only essential whorls

(4) All flowering plants produce flowers

59) A biologist observes a flower with brightly coloured petals, sticky pollen grains. The pollen grains germinate easily when placed on the stigma of another flower of the same species. The most probable pollination agent is:

(1) Wind

(2) Water

(3) Insects

(4) Either wind or insect

60) What is the ploidy level of Nucellus, MMC, functional megaspore and female gametophyte ?

(1) 2n, n, 2n, 2n

(2) 2n, n, 2n, n

(3) 2n, 2n, n, n

(4) n, 2n, n, n

61) Read the following four statements carefully:
I. Unicellular organisms are incapable of independent existence because a single cell cannot manage all life processes.
II. Cell theory in its modern form includes the principle that all living organisms are composed of cells and products of cells.
III. Robert Hooke discovered the cell wall in 1665, which was actually a dead structure.
IV. A.V. Leeuwenhoek discovered the nucleus in 1674 while observing living bacterial cells.
Select the option containing the combination of incorrect statements:

(1) II and III

(2) I and IV

(3) II and IV

(4) III and IV

62)

Choose the incorrect statement :

(1) Onion cells possess a distinct cell wall as their outer boundary, inside of which lies the cell membrane.

(2) Human cheek cells lack a cell wall and have an outer membrane as their delimiting structure.

(3) Prokaryotic cells contain non-membrane bound organelles like ribosomes and centrosomes.

(4) Mycoplasmas are the smallest cells, measuring approximately 0.3 µm in length.

63) Read the following statements :
I. The nucleus contains chromosomes, which hold the genetic material, DNA.
II. Non-membrane bound ribosomes are strictly absent in prokaryotic cells.
III. The largest isolated single cell is the egg of an ostrich.
IV. Centrosome is a non-membrane bound organelle found in animal cells that aids in cell division.
Select the option containing the combination of correct statements:

(1) I, III, and IV

(2) I, II, and III

(3) II and IV

(4) All I, II, III, and IV

64) Match Column-I (Cell/Structure) with Column-II (Characteristic/Dimensions):

Column-I	Column-II
A. Mycoplasma	1. 3 to 5 µm in length
B. Human RBCs	2. 10 to 20 µm in size
C. Bacteria	3. 0.3 µm in length
D. Eukaryotic cell	4. About 7.0 µm in diameter

(1) A-1, B-4, C-3, D-2

(2) A-3, B-1, C-4, D-2

(3) A-4, B-3, C-1, D-2

(4) A-3, B-4, C-1, D-2

65) **Assertion (A):** The presence of a nuclear envelope is an important feature distinguishing eukaryotic organisation from the prokaryotic organisation.
Reason (R): Eukaryotic cells possess an organised nucleus with a nuclear envelope, whereas prokaryotic genetic material is not enveloped by a nuclear membrane.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

66) Which combination contains only features specifically associated with plant cells ?

(1) Centrioles + plastids + cell wall

(2) Cell wall + plastids + large central vacuole

(3) Centrioles + large central vacuole + plastids

(4) Nuclear envelope + chromosomes + centrioles

67) Regarding plasma membrane which option gives the number of correct statements?
A. Its major lipids are phospholipids.
B. Phospholipids are arranged as a bilayer.
C. Polar heads face the outer sides of the bilayer.
D. Hydrophobic tails face the inner part.
E. Cholesterol is also present in the membrane.

(1) Two

(2) Three

(3) Four

(4) Five

68) **Assertion (A):** The phospholipid arrangement in the plasma membrane helps protect the nonpolar hydrocarbon tails from the aqueous environment.
Reason (R): The polar heads face the outer sides, while the hydrophobic tails are directed towards the inner part of the bilayer.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

69) **Statement I :** Fluidity refers to the ability of membrane components, particularly proteins, to move laterally within the overall bilayer.
Statement II : The fluid nature of the membrane has no role in secretion or cell division.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

70) A polar molecule needs to cross the plasma membrane. Direct passage through the nonpolar lipid bilayer is not possible for this molecule.
Which mechanism would facilitate its movement?

(1) Passage directly through phospholipid tails

(2) Carrier protein-mediated transport

(3) Movement exclusively through cholesterol

(4) Movement through the hydrophobic region without assistance

71) Which organelle is not included in the endomembrane system ?

(1) Endoplasmic reticulum

(2) Golgi complex

(3) Lysosome

(4) Mitochondrion

72) **Assertion (A):** Cells synthesizing steroidal hormones possess prominent SER.
Reason (R): SER is the major site for synthesis of lipids, and steroidal hormones.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

73) A cell contains a membrane-bound compartment with water, cell sap, excretory products and other materials not useful for the cell. The compartment is surrounded by a single membrane. Which combination correctly identifies the structure and its membrane?

(1) Lysosome — plasma membrane

(2) Vacuole — tonoplast

(3) Vacuole — cristae

(4) ER — tonoplast

74) Consider the following observations:
P. A plant cell contains a vacuole occupying a major fraction of the cell volume.
Q. The membrane surrounding this vacuole is called tonoplast.
R. The tonoplast can facilitate movement of ions against their concentration gradients into the vacuole.
S. Therefore, ion concentration inside the vacuole can become significantly higher than in the cytoplasm.
Which option is correct?

(1) P and Q only

(2) P, Q and R only

(3) Q, R and S only

(4) P, Q, R and S

75)

Consider the following statements about the axoneme:
A. It is the core of a cilium or flagellum.
B. It possesses microtubules running parallel to the long axis.
C. It usually contains nine peripheral doublets and two centrally located microtubules.

D. The arrangement is referred to as the 9+2 array.
E. The axoneme is not covered by the plasma membrane.
How many statements are correct?

(1) Two

(2) Three

(3) Four

(4) Five

76) Consider the following statements regarding the connections within an axoneme:
A. The central tubules are connected by bridges.
B. The central tubules are enclosed by a central sheath.
C. The central sheath is connected to one tubule of each peripheral doublet by a radial spoke.
D. There are nine radial spokes.
Mark the option with correct statements.

(1) A and B only

(2) A, B and C only

(3) B, C and D only

(4) A, B, C and D

77) Read the given statements :
A. Chromatin is a nucleoprotein material.
B. Chromatin contains DNA, histones, non-histone proteins and RNA.
C. During cell division, structured chromosomes appear in place of the interphase nucleus.
D. Every chromosome is visible only in dividing cells.
E. A single human cell contains approximately two metres of DNA distributed among 46 chromosomes.
Which option correctly represents the number of correct statements?

(1) Two

(2) Three

(3) Four

(4) Five

78) Four chromosomes have the following centromere positions:
Chromosome P: Exactly at the middle, producing two arms of equal length.
Chromosome Q: Close to one end, producing one extremely short and one very long arm.
Chromosome R: Slightly away from the middle, producing one shorter and one longer arm.
Chromosome S: At the terminal position.
Which sequence correctly identifies P, Q, R and S?

(1) Metacentric → Sub-metacentric → Acrocentric → Telocentric

(2) Sub-metacentric → Metacentric → Telocentric → Acrocentric

(3) Metacentric → Acrocentric → Sub-metacentric → Telocentric

(4) Telocentric → Sub-metacentric → Acrocentric → Metacentric

79) The following observations are made from a eukaryotic cell:

I. Nuclear envelope consists of two parallel membranes.
II. The space between these membranes is called the perinuclear space.
III. The outer nuclear membrane is usually continuous with the ER and bears ribosomes.
IV. Nuclear pores are formed by fusion of the two nuclear envelope membranes.
V. RNA and protein molecules can move through nuclear pores only in one direction i.e., from nucleus to cytoplasm.
Which option gives the maximum number of correct observations ?

(1) Three

(2) Four

(3) Five

(4) None

80) A student compares the nuclear envelope, ER and nucleolus and makes the following claims :
Choose the incorrect statement?

(1) Secondary constriction region organises nucleolus.

(2) The nucleolus is a membrane-bound structure enclosed within the nucleoplasm.

(3) The nucleolus is involved in active ribosomal RNA synthesis.

(4) The nucleoplasm contains chromatin and nucleolus.

81) A student compares three structures — **mitochondria, chloroplast and ribosomes** — and records the following observations:
Which of the following observations is incorrect?

(1) One is associated with cellular respiration and possesses its own genetic material and ribosomes.

(2) One is associated with photosynthetic activity and possesses its own genetic material and ribosomes.

(3) One is the site of protein synthesis and occurs in both prokaryotic and eukaryotic cells.

(4) The first two possess a double membrane, whereas the third possesses two membrane-bound subunits respectively.

82) Which of the following combinations is incorrectly matched ?

(1) SER — Usually contains tubules and lacks ribosomes

(2) Golgi apparatus — Modification and packaging of materials; formation of glycoproteins and glycolipids

(3) Lysosome — Hydrolytic digestion at acidic pH; formed by packaging in Golgi apparatus

(4) Vacuole — Possesses hydrolytic enzymes active at pH of 5.

83) **Assertion (A):** Mycoplasma is an exception to the general presence of a cell wall in prokaryotic cells.
Reason (R): Most prokaryotic cells have a cell wall surrounding the plasma membrane, whereas mycoplasma lacks a cell wall.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

84) Match Column I with Column II:

Column I	Column II
A. Glycocalyx	P. Determines bacterial shape and provides structural support
B. Cell wall	Q. Selectively permeable and interacts with the outside world
C. Plasma membrane	R. May occur as slime layer or capsule
D. Plasmid	S. Small circular DNA outside genomic DNA

(1) A-P, B-Q, C-R, D-S

(2) A-R, B-Q, C-P, D-S

(3) A-R, B-P, C-Q, D-S

(4) A-S, B-P, C-Q, D-R

85) Which of the following is **incorrect** regarding transport across the plasma membrane?

(1) Simple diffusion of neutral solutes occurs along the concentration gradient.

(2) Osmosis refers to movement of water by diffusion across a membrane.

(3) Active transport can move ions or molecules against their concentration gradient.

(4) Active transport occurs without energy expenditure because movement is directed down the concentration gradient.

86) A bacterial cell is found to possess a **loose outer sheath**. Another bacterial cell possesses a **thick and tough outer covering** along with the same underlying structures. Which pair correctly identifies the two outer coverings?

(1) Capsule and slime layer, respectively

(2) Slime layer and capsule, respectively

(3) Cell wall and capsule, respectively

(4) Plasma membrane and slime layer, respectively

87) **Statement I:** Non-polar molecules generally require carrier channel proteins for transport across the plasma membrane.
Statement II: Non-polar molecules readily pass through the nonpolar lipid bilayer by simple diffusion.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct, but Statement II is incorrect.

(4) Statement I is incorrect, but Statement II is correct.

88) Consider the following structures/features:
A. Phosphate granules — inclusion body
B. Glycogen granules — inclusion body

C. Gas vacuoles — found in some blue-green and photosynthetic bacteria
D. Chromatophores — membranous extensions containing pigments in some prokaryotes
E. Fimbriae — structures primarily responsible for bacterial motility
How many of the above are correctly matched?

(1) Two

(2) Three

(3) Four

(4) Five

89) Which of the following best describes a glyoxysome?

(1) A membrane-bound minute vesicle belonging to the group of microbodies.

(2) A non-membranous structure involved in protein synthesis.

(3) A specialised infolding of plasma membrane found exclusively in bacteria.

(4) A double-membrane organelle responsible for cellular respiration.

90) A bacterial cell is observed to have the following characteristics:

A. a circular genomic DNA molecule
B. additional small circular DNA
C. 70S ribosomes
D. no membrane-bound nucleus
E. inclusion bodies free in the cytoplasm
F. surface appendages that do not participate in motility
G. and a selectively permeable plasma membrane.

Which combination correctly identifies the structures represented by the additional DNA, ribosomes and non-motile surface appendages, respectively?

(1) Plasmid, 70S ribosome, flagella

(2) Plasmid, 70S ribosome, pili/fimbriae

(3) Plasmid, 80S ribosome, flagella

(4) Chromatin, 80S ribosome, pili

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	2	4	1	2	1	4	3	3	4	4	4	2	1	2	4	4	3	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	2	1	3	1	1	3	3	2	1	1	1	3	2	1	3	2	1	2
Q.	41	42	43	44	45															
A.	3	3	2	4	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	1	2	1	2	3	4	1	1	1	2	1	1	3	2	2	4	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	4	3	1	3	4	3	4	2	3	2	1	3	4	3	2	2	3	3
Q.	86	87	88	89	90															
A.	4	1	3	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	3	4	1	2	2	1	1	4	2	3	1	4	2	1	4	1	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	3	2	3	2	3	1	2	1	2	3	3	4	4	2	1	1	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	1	4	1	1	3	2	3	4	3	4	3	1	3	1	2	3	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	3	1	4	1	2	4	1	3	2	4	1	2	4	3	4	4	3	2	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	1	3	4	2	4	3	1	2										

SOLUTIONS

PHYSICS

1)

Question Explanation:

The question asks for the physical quantity whose dimensions are the same as $\frac{A^3}{B}$ given the position function $x(t) = \frac{A}{B}\left(1 - e^{-At}\right)$.

Concept:

Dimensional Analysis

The argument of an exponential function must be dimensionless, which helps determine the dimensions of constants.

Solution:

Given:

$x(t) = \frac{A}{B}\left(1 - e^{-At}\right)$

Since x represents position, its dimension is length [L].

The exponent –A t must be dimensionless:

$[A][t] = 1 \Rightarrow [A] = T^{-1}$

Now,

$x = \frac{A}{B}$

So,

$\frac{A}{B} = L$

$[B] = \frac{A}{L} = \frac{T^{-1}}{L}$

Now find dimensions of $\frac{A^3}{B}$

$\frac{\left(T^{-1}\right)^3}{T^{-1}/L} = \frac{T^{-3}}{T^{-1}} \times L = LT^{-2}$

This is the dimension of acceleration.

Final Answer:

Option (4)

2) **Explain question :** To find the average speed for total distance travelled by car.

Concept : This question is based on speed and velocity (average and instantaneous)

Solution :

$v_{avg} = \frac{d}{\frac{2d}{5v_1} + \frac{3d}{5v_2}}$

$v_{avg} = \frac{5v_1v_2}{2v_2 + 3v_1}$

Final answer : 4

3) **Explaining :** We compute acceleration using the change in velocity over time.
Concept : This question is based on definition of average acceleration.
Solution :
$v_i = 10 + 2(2)^2 = 18$ m/s
$v_f = 10 + 2(5)^2 = 60$ m/s
$a_{av} = \frac{\Delta v}{\Delta t} = \frac{60-18}{3} = 14$ m/s^2
Final Answer : (4)

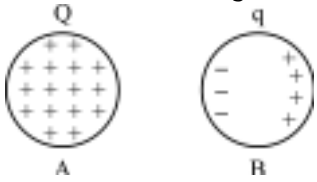
4) **Explain the question :** In this question we are given with the relation between distance 'x' and time 't'. We need to find the retardation of the particle.
Concept : This question is based on application of calculus for general motion.
Solution : Given here
$t = \alpha x^2 + \beta x$ (α and β are constant)
Differentiation with respect to 'x'.
$\Rightarrow \frac{dt}{dx} = \frac{d}{dx}(\alpha x^2 + \beta x)$
$\Rightarrow \frac{dt}{dx} = 2\alpha x + \beta \Rightarrow \frac{dx}{dt} = \frac{1}{2\alpha x + \beta}$
 velocity, $v = \frac{dx}{dt} = \frac{1}{2\alpha x + \beta}$ ....(2)
acceleration, $a = \frac{cdc}{dx}$
$a = \frac{1}{2\alpha x + \beta}\frac{d}{dx}(2\alpha x + \beta)^{-1}$
$a = \frac{1}{2\alpha x + \beta}\left(\frac{-2\alpha}{(2\alpha x + \beta)^2)}\right) = \frac{-2\alpha}{(2\alpha x + \beta)^3}$ (negative sing shows deceleration)
 $\because 2\alpha x + \beta = \frac{1}{V}$ (from equation (1)
 $a = 2\alpha v^3$ (considering the magnitude only)
Final answer : 1

5) **Question Explanation:** To find tension using equilibrium condition of charges.
Concept: This question is based on Coulomb's law and equilibrium of charges
Solution:
$F = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$
T sin θ = F, T cos θ = mg
Geometry:
$\sin\theta = \frac{a/2}{\ell} = \frac{a}{2\ell}$
From horizontal balance: $T\sin\theta = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$ Substitute $\sin\theta = \frac{a}{2\ell}$:
$T\sin\theta = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$
$T \cdot \frac{a}{2\ell} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$

$$T = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}\cdot\frac{2\ell}{a}$$
$$T = \frac{q^2\ell}{2\pi\varepsilon_0 a^3}$$
Final Answer: (2)

6) **Explaining :** To find the magnitude of force when charges are placed at corners of cube of side length b.
Concept : This question is based on coulombs law
Solution :
$$F\propto\frac{1}{a^2}\quad F_1\propto\frac{1}{b^2}\Rightarrow F_1=\frac{Fa^2}{b^2}$$
Final Answer : (1)

7) **Explaining :** To determine the nature of force between similarly charged bodies.
Concept : This question is based on Coulomb's law and induction.
Solution :
Two similar charged bodies may attract each other only in special case.

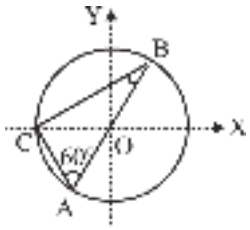


If Q and q are similar charge and induced charge of body B is at smaller distance to A as compare to positive charge of B. If q < Q there there is a possibility that attraction is more than repulsion.
Final Answer : (4)

8)

QUESTION EXPLANATION: To find the maximum velocity of the particle.
CONCEPT: This question is based on kinematics of motion
SOLUTION:
$$v(t) = \frac{dx}{dt} = \frac{d}{dt}(18t - 2t^3 + 6t + 5) = 36t - 6t^2 + 6$$
$$a(t) = \frac{dv}{dt} = \frac{d}{dt}(36t - 6t^2 + 6) = 36 - 12t.$$
$36 - 12t = 0 \Rightarrow 12t = 36 \Rightarrow t = 3s.$
$v_{max} = v(3) = 36(3) - 6(3)^2 + 6 = 60$
Final Answer: (3)

9) **Explaining:** To find electric field, potential, potential energy and force between the charges in the given system
Concept: This question is based on electrostatics of charges
Solution:
Electric field at point O is vector sum of electric field due to all three point charges individually.



$\vec{E}_O = \vec{E}_A + \vec{E}_B + \vec{E}_C$

and $\vec{E}_A = -\vec{E}_B$

So, $\vec{E}_O = \vec{E}_C$

$= \frac{2q/3}{4\pi\varepsilon_0 R^2}$ along $\vec{OC} = \frac{q}{6\pi\varepsilon_0^2}$ along $\vec{OC}$

$= -\frac{q}{6\pi\varepsilon_0 R^2}\hat{i}$

From geometry, we can find that ∠ABC = 30° and ∠ACB = 90°

So AB = 2R; AC = R; BC = $\sqrt{3}R$

Total potential energy of the system is,

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{q/3 \times q/3}{2R} - \frac{q/3 \times 2q/3}{R} - \frac{q/3 \times 2q/3}{\sqrt{3}R}\right] \neq 0$$

Force of interaction between B and C is,

$$F = \frac{q/3 \times 2q/3}{4\pi\varepsilon_0 \times (\sqrt{3}R)^2} = \frac{q^2}{54\pi\varepsilon_0 R^2}$$

Potential at point O is , $V = \frac{q/3 + q/3 - 2q/3}{4\pi\varepsilon_0 R} = 0$

Final Answer: (3)

10) **Explaining:** To find position of equilibrium of a charge between two positive charges

Concept: This question is based on force between the charges

Solution:

if q is + ve

axial displacement along AB → stable equilibrium equational displacement along CD → unstable equilibrium

if q is – ve

axial displacement along AB → unstable equilibrium equational displacement along CD → stable equilibrium

Hence all of the above

Final Answer: (4)

11) **Explaining:** To find point at which electric field intensity is zero

Concept: This question is based on electric field due to point charges

Solution:

For E = 0, $E_1 = E_2$

$$\frac{KQ}{x^2} = \frac{2KQ}{(x+3)^2}$$

$$\Rightarrow \sqrt{2}x = x + 3 \Rightarrow x = \frac{3}{\left(\sqrt{2}-1\right)} = 3\left(\sqrt{2}+1\right) = \left(3\sqrt{2}+3\right)$$

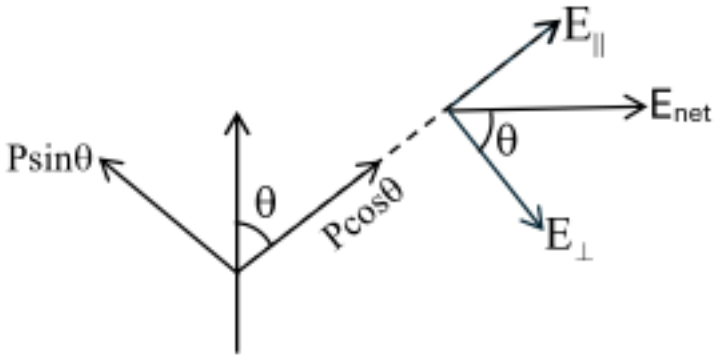
from 2Q ⇒ x + 3 = $\left(3\sqrt{2}+3\right) + 3 = \left(3\sqrt{2}+6\right)$

Final Answer: (4)

12) **Question Explanation:**
The question asks to find the angle θ between the electric dipole $\vec{P}$ and the position vector of point A, such that the resulting net electric field at A is perpendicular to $\vec{P}$.

Concept:
Electric Field at a General Point due a dipole

Solution :



$$\tan\theta = \frac{E_{\parallel}}{E_{\perp}} = \frac{2k\vec{P}\cos\theta}{\frac{r^3}{\frac{kP\sin\theta}{r^3}}}$$

$$\tan\theta = \frac{2\cos\theta}{\sin\theta}$$

$$\tan^2\theta = 2$$

$$\tan\theta = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\sqrt{2}\right)$$

Final Answer: Correct Option is (4)

13) **Explaining :** Find unit vector in direction of friction force acting on block.
Concept : Direction of friction will be opposite to the direction of velocity of block with respect to plank.

$$\hat{f} = -\hat{v}_{rel} = -\hat{v}_{AP} = -\left[\frac{4\hat{j} - 3\hat{i}}{5}\right] = \frac{3\hat{i} - 4\hat{j}}{5}$$

Solution :
Final Answer : (2)

14) **Explaining:** To find direction of cosine of the vector
Concept: This question is based on cosine law
Solution:

$$\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$$

$$\left|\vec{A}\right| = \sqrt{(2)^2 + (4)^2 + (-5)^2} = \sqrt{45}$$

$$\therefore \quad \cos\alpha = \frac{2}{\sqrt{45}}, \cos\beta = \frac{4}{\sqrt{45}}, \cos\gamma = \frac{-5}{\sqrt{45}},$$

Final Answer: (1)

15) **Explaining :** We need to find projection (component) of $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ on $\vec{B} = -\hat{i} + 3\hat{j} + 4\hat{k}$

Concept : This question is based on projection of $\vec{A}$ on $\vec{B}$ is written as $\frac{\vec{A}.\vec{B}}{B}$ in scalar form.

Solution :

As,

$$\left|\vec{A}\right| = \sqrt{(2)^2 + (3)^2 + (-1)^2} = \sqrt{4 + 9 + 1} = \sqrt{14}$$

and

$$\left|\vec{B}\right| = \sqrt{(-1)^2 + 3^2 + 4^2} = \sqrt{1 + 9 + 16} = \sqrt{26}$$

$$\therefore \vec{A}.\vec{B} = 2(-1) + 3 \times 3 + (-1)(4) = 3$$

The projection of A on B (i.e., A cosθ) = $\frac{\vec{A}.\vec{B}}{|B|} = \frac{3}{\sqrt{26}}$

Final Answer : (2)

16) **Explaining:** To find value of a for the given vector

Concept: This question is based on unit vector

Solution:

$$\hat{A} = a\hat{i} + 0.8\hat{j}$$

$$\left|\hat{A}\right| = 1$$

So, $a^2 + (0.8)^2 = 1^2$

$a^2 = 0.36$

$a = \pm 0.6$

Final Answer: (4)

17) **Explaining :** To find electric field and flux due to charges

Concept : This question is based on Gauss theorem.

Solution : $Q_{en} = 0$ so ϕ must be zero.

Final Answer : (4)

18) **Explaining :** A charge – q of mass m is rotating in a circle of radius r around an infinitely large uniformly charged wire as shown

Concept : This question is based on $\frac{mv^2}{r} = qE$

Solution :

$$F_e = \frac{mv^2}{r} \Rightarrow \frac{\lambda q}{2\pi\varepsilon_0 r} = \frac{mv^2}{r}$$

$\Rightarrow$ v is independent of r.

Final Answer : (3)

19)

QUESTION EXPLANATION

The objective is to find the value of the integral $\int \left(\frac{\cos^2 x - \sin^2 x}{\sec^2 x - \tan^2 x}\right) dx$.

CONCEPT
This problem requires the use of two fundamental trigonometric identities:
The double angle identity for cosine: $\cos(2x) = \cos^2x - \sin^2x$.
The Pythagorean identity involving secant and tangent: $\sec^2x - \tan^2x = 1$.

SOLUTION
1. First, simplify the expression inside the integral.
2. The numerator is the double angle identity for cosine: $\cos^2x - \sin^2x = \cos(2x)$.
3. The denominator is a Pythagorean identity: $\sec^2x - \tan^2x = 1$.
4. Substitute these identities back into the integral:
$$\int \left(\frac{\cos(2x)}{1}\right) dx = \int \cos(2x)dx$$
.
5. Now, evaluate the integral of cos(2x). This is a standard integral form:
$$\int \cos(ax)dx = \frac{\sin(ax)}{a} + C$$
.
6. Applying this with a = 2, we get:
$$\int \cos(2x)dx = \frac{\sin(2x)}{2} + C$$
.

CORRECT ANSWER
The correct option is (3).

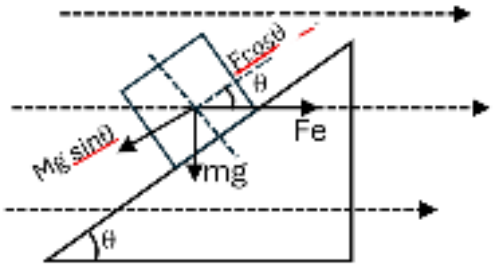
20) **QUESTION EXPLANATION**
The question asks to use the binomial approximation $(1 + x)^n \approx 1 + nx$ (for small |x|) to find the approximate value of $(999)^{1/3}$.

CONCEPT
The binomial approximation is a powerful tool for estimating roots and powers of numbers that are very close to a number whose root/power is easy to calculate (like 1, 100, 1000).
The key is to rewrite the number in the form of (A + B), where B is much smaller than A, and then factor out A to get it into the $(a + x)^n$ form.

SOLUTION
1. First, rewrite 999 as a number close to a perfect cube. The closest perfect cube is 1000.
999 = 1000 – 1
2. So, we need to calculate $(1000 - 1)^{1/3}$.
3. Factor out 1000 to get the expression into the $(1 + x)^n$ form:
$$\left(1000\left(1-\frac{1}{1000}\right)\right)^{1/3} = (1000)^{1/3} \cdot \left(1-\frac{1}{1000}\right)^{1/3}$$
4. $(1000)^{1/3} = 10$.
5. Now apply the binomial approximation to the second part, $\left(1-\frac{1}{1000}\right)^{1/3}$.
Here, x = –1/1000 and n = 1/3. Since |x| is very small, the approximation is valid.
$$\left(1-\frac{1}{1000}\right)^{1/3} \approx 1 + nx = 1 + \left(\frac{1}{3}\right)\left(-\frac{1}{1000}\right) = 1 - \frac{1}{3000}.$$
6. Combine the parts to get the final approximation:
$$10 \cdot \left(1-\frac{1}{3000}\right) = 10 - \frac{10}{3000} = 10 - \frac{1}{300}.$$

CORRECT ANSWER
The correct option is (1).

21) **Explaining :** The question is asking about θ when block under equilibrium. **Concept :** This question is based on electrostatic force i.e. force on block due to electric field is F = qE.
Solution :
Formula : F = qE
Visual -Aid:-



Solution / Calculation
For equilibrium

$$mg\sin\theta = F\cos\theta$$

$$\frac{\sin\theta}{\cos\theta} = \frac{F}{mg}$$

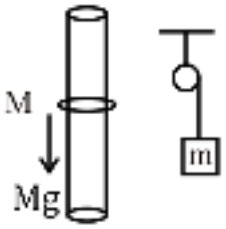
$$\tan\theta = \frac{20\times 2}{10\times 10} = \frac{4}{10} = \frac{2}{5}$$

$$\theta = \tan^{-1}\left(\frac{2}{5}\right)$$

Conclusion : Hence, option (2) is correct.
Final Answer : (2)

22) **Question Explanation :**
Find the frictional force acting on a ring of mass M kept in equilibrium on a vertical pole, connected to a hanging block m.

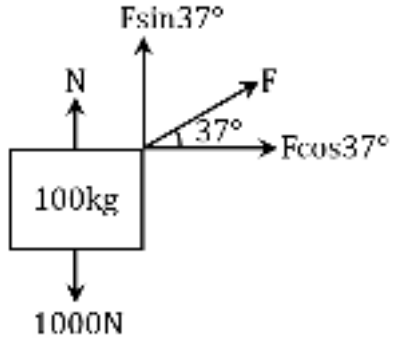
Concept:
Static Equilibrium (ΣF = 0)



Solution:
$f_{max} = \mu N$
$f_{max} = \mu mg$
friction required = Mg

Final Answer: Option (1)

23) **Question Explanation:** Find the maximum acceleration of a boy climbing a rope such that the attached block remains stationary.
Concept: Newton's Laws & Friction ($f_s \le \mu N$)
Solution:



$F\cos 37° = \mu\left(1000 - \frac{3F}{5}\right)$

$\frac{4}{5}F = \frac{1000}{3} - \frac{F}{5}$

$F = \frac{1000}{3}$

$\frac{1000}{3} = 25\,(g + a)$

$\frac{40}{3} = (g + a)$

$\frac{4g}{3} = g + a$

$\frac{g}{3} = a$

Final Answer: Option (2)

24) **Question Explanation:** For the coplanar concurrent forces shown acting on an object to keep it in the state of equilibrium. we need to find $|\vec{F}_1|$ and $|\vec{F}_2|$.

Concept : Translational equilibrium.

Solution :

Horizontal equilibrium : $8 + 4\cos 60° = F_2 \cos 30° \Rightarrow F_2 = \frac{20}{\sqrt{3}}N$

Vertical equilibrium : $F_1 + 4\cos 30° = F_2 \sin 30° \Rightarrow F_1 = \frac{4}{\sqrt{3}}N$

Final Answer : Option (1)

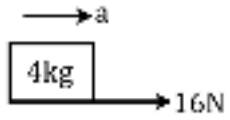
25) **Question Explanation:** For the two blocks system shown in the figure, we need to find maximum value of force F (as indicated) such that both the blocks move with same acceleration.

Concept: Two blocks system in friction (on smooth surface)

Solution:

$a_{sys} = \frac{F}{12}$

$f_L = 0.4 \times 40 = 16N$



$16 = 4 \times \frac{F}{12}$

F = 48 N

Final answer: option (3)

26) **Question Explanation:** The question asks what each spring balance will read.
Concept: In series combination reading of each spring balance is same.
Solution:
 spring is massless
Reading = T = 10 kg–wt
Final Answer: Option (1)

27)

Explaining : Find force by clamp on pulley.
Concept : $F_{net} = ma$
and $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$
Solution : $a_{sys} = \frac{mg}{2(2m)} = \frac{g}{4}$

$T = m(g-a) = \frac{3mg}{4}$

$R = 2T\cos\frac{60^\circ}{2} = 2T \times \frac{\sqrt{3}}{2}$

$R = \sqrt{3}T = \sqrt{3}\left(\frac{3mg}{4}\right)$

Final Answer : (1)

28) **Question Explanation:** Find the change in momentum of body acted upon by a variable force shown in the F – t graph.
Concept: Impulse-Momentum Theorem (I = ΔP) and Impulse is the area under the F – t curve.
Solution: The momentum acquired (ΔP) is equal to the area under the F – t graph from t = 0 to t = 9s.
Calculation: Area = $\frac{1}{2} \times 4 \times 10 + 5 \times 10 = 70$ N–s
Final Answer : Option (3)

29) **Explaining :** find acceleration at x = 8m
Concept : This question is based on $a = v\frac{dv}{dx}$
Solution :

$$a = x^{\frac{2}{3}} \frac{dx^{\frac{2}{3}}}{dx}$$
$$a = x^{\frac{2}{3}} . \frac{2}{3} x^{\frac{2}{3}-1}$$
$$a = x^{\frac{2}{3}} . \frac{2}{3} x^{-\frac{1}{3}}$$
$$a = \frac{2}{3} x^{\frac{1}{3}} = \frac{4}{3} \text{ m/s}^2$$
Final Answer : (3)

30) **Explanation:** To find force on a dipole in electric field
Concept: This question based on force $F = p\frac{dE}{dx}$ **Solution:**
$\vec{F} = p\frac{dE}{dx}\hat{i}$. Here $\frac{dE}{dx} = \alpha$, so $\vec{F} = p\alpha\hat{i}$
Final Answer: Option (2)

31) **Explaining :** Figure shows the field lines of a positive and negative point charge respectively
Concept : This question is based on $\Delta u = q\Delta V$
Solution :
$$U_P - U_Q = \underbrace{-q}_{-ive} \underbrace{[V_P - V_Q]}_{+ive}$$
$$= -ive$$
$$U_B - U_A = \underbrace{-q}_{-ive} \underbrace{[V_B - V_A]}_{+ive}$$
$$= -ive$$
Final Answer : (1)

32) **Explaining :** When adding or subtracting experimental measurements, the precision of the final answer is limited by the least precise measurement involved—specifically, the one with the fewest decimal places.
Concept : This question is based on Significant Figures: Rules for Addition and Subtraction.
Solution :
Line up the values by their decimal points to check precision:
12.340 g (precise to 3 decimal places)
0.56 g (precise to 2 decimal places)
105.2 g (precise to 1 decimal place)
Raw sum: 12.340 + 0.56 + 105.2 = 118.100 g
Since 105.2 g is only precise to one decimal place, the final answer must be rounded to exactly one decimal place: 118.1 g.
Final Answer : (1)

33) **Explaining :** choose correct graph
Concept : This question is based on v-h curve
Solution :

$(v^2) \propto (\pm h)$

Final Answer : (1)

34) **Explaining :** Find θ

Concept : This question is based on projectile motion

Solution :

From the given diagram,

$$\frac{R/2}{H} = \frac{\sqrt{3}H}{H} = \sqrt{3}$$

or $$\frac{(v_0^2 \sin\theta\cos\theta)/g}{(v_0^2\sin^2\theta)/2g} = \sqrt{3} \Rightarrow 2\cot\theta = \sqrt{3}$$

or $$\tan\theta = \frac{2}{\sqrt{3}}$$

or $$\theta = \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

Final Answer : (3)

35) **Explaining :** Match the column

Concept : This question is based on equation of trajectory

Solution :

Compare with equation of trajectory

$$y = x(\tan\theta) - \left(\frac{g}{24^2\cos^2\theta}\right) \times 2$$

$$\tan\theta = 1 \Rightarrow \theta = 45^\circ$$

and $$\frac{g}{2u^2\cos^2\theta} = \frac{1}{80} \Rightarrow \frac{10}{2u^2\left(\frac{1}{2}\right)^2} = \frac{1}{80}$$

$$u = 20\sqrt{2}\,m/sec$$

$$H_{max} = \frac{u^2\sin^2\theta}{2g} = \frac{800\left(\frac{1}{\sqrt{2}}\right)^2}{20} \Rightarrow H_{max} = 20m$$

Range, $$R = \frac{u^2\sin^2\theta}{2g} \Rightarrow R = \frac{800\sin 90^\circ}{10} = 80m$$

After 4 sec

$u_x = 20\sqrt{2}\cos 45^\circ$ (constant)

u_x = 20 m/sec

and vertical velocity

$v_y = u_y + a_yt$

$= 20\sqrt{2}\sin 45^\circ - g(4)$

v_y = -20 m/sec

$$\tan\theta = \frac{v_y}{v_x} = -\left(\frac{20}{20}\right) \Rightarrow \tan\theta = -1$$

$\theta = -45^\circ$ (-ve means is downward)

Final Answer : (2)

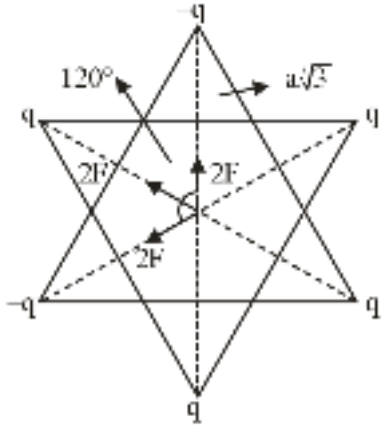
36) **Explaining :** Find magnitude of relative velocity of the boat with respect to water

Concept : This question is based on river boat concept

Solution :

$\vec{V}_{BR} = \vec{V}_B - \vec{V}_R$
$\vec{V}_{BR} = 3\hat{i} + 4\hat{j} + 3\hat{i} + 4\hat{j}$
$\vec{V}_{BR} = 6\hat{i} + 8\hat{j}$
$\vec{V}_{BR} = 10 m/s$
Final Answer : (1)

37) **Explaining :** We are asked to determine net force on q.
Concept : This question is based on resultant of forces.
Solution :



$F_{net} = 2F + 2F = 4F$

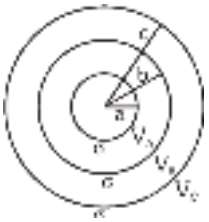
$= \dfrac{12kq^2}{a^2} \qquad \left(F = \dfrac{kq^2}{\left(a/\sqrt{3}\right)^2} \right)$

Final Answer : (3)

38) **Explaining :** To find the relation b/w potential of given shells.

$$V = \begin{cases} \dfrac{KQ}{R} = \dfrac{\sigma R}{\varepsilon_0} & ; \quad r \leqslant R \\ \dfrac{KQ}{r} = \dfrac{\sigma R^2}{\varepsilon_0 r} & ; \quad r > R \end{cases}$$

Concept : This question is based on
Solution :



for $c = a + b \Rightarrow V_C = V_A \neq V_B$

$V_A = \dfrac{\sigma a}{\varepsilon_0} - \dfrac{\sigma b}{\varepsilon_0} + \dfrac{\sigma c}{\varepsilon_0}$

$V_B = \dfrac{\sigma a^2}{\varepsilon_0 b} - \dfrac{\sigma b}{\varepsilon_0} + \dfrac{\sigma c}{\varepsilon}$

$V_C = \dfrac{\sigma a^2}{\varepsilon_0 c} - \dfrac{\sigma b^2}{\varepsilon_0 c} + \dfrac{\sigma c}{\varepsilon_0}$

$\because c = a + b \Rightarrow V_A = V_C$

Final Answer : (2)

39) **Explaining :** To find the magnitude of the electric field at the point (1,3,1).
Concept : This question is based on
The electric field $\vec{E}$ is the **negative gradient** of the potential function:

$$\vec{E} = -\left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}\right) = E_x\hat{i} + E_y\hat{j} + E_z\hat{z}$$

Solution : Given:
$\phi(x, y, z) = x^2y + yz$
Compute partial derivatives:

$$\frac{\partial\phi}{\partial x} = 2xy \quad \frac{\partial\phi}{\partial y} = x^2 + z \quad \frac{\partial\phi}{\partial z} = y$$

At point (1, 3, 1):

$$\frac{\partial\phi}{\partial x} = 2(1)(3) - 6 \Rightarrow E_x = -6$$

$$\frac{\partial\phi}{\partial y} = 1^2 + 1 = 2 \Rightarrow E_y = -2$$

$$\frac{\partial\phi}{\partial z} = 3 \Rightarrow E_z = -3$$

Now calculate magnitude:

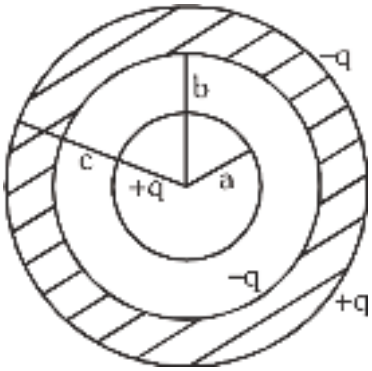
$$\left|\vec{E}\right| = \sqrt{(-6)^2 + (-2)^2 + (-3)^2}$$
$$= \sqrt{36 + 4 + 9} = \sqrt{49} = 7$$

Final Answer : (1)

40) **Explaination :** To find electric field due to charged sphere.
Concept : This question is based on electric field inside a uniformly charged solid sphere (non-conducting).
Solution : Let's verify each statement.
(i) The electric field for r < a:
This is **inside** the uniformly charged non-conducting sphere:

$$E = \frac{1}{4\pi\epsilon_0}\frac{qr}{a^3} \rightarrow \text{True}$$

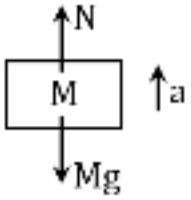
(ii) Electric field for a < r < b :
This region lies **outside the solid sphere** but **inside the cavity** of the conductor (no charge here).
Using **Gauss's law**, the field is due to charge +q enclosed:
$E \neq 0 \rightarrow$ FALSE
(iii) Electric field for b < r < c :



This is **inside the conductor**, and electrostatic field inside a conductor is always:
$E = 0 \rightarrow$ TRUE
(iv) Charge on the inner surface of the conducting shell:

Since a **+ q** charge is inside the shell, it **induces -q** on the inner surface.
→ TRUE
(v) Charge on the outer surface of the shell:
Net charge on shell = −q, and **inner surface has -q**, so outer surface must have zero charge.
→ FALSE
Final Answer : (2)

41) **Explaining :** Find acceleration of box.
Concept : This question is based on F_{net} = ma.



Solution :
FBD of M : If M exerts force N = 7 Mg/4 on floor, then from third law floor also exerts force N on box in upward direction.
N – Mg = Ma
$\Rightarrow \frac{7\,Mg}{4} - Mg = Ma \Rightarrow a = \frac{3g}{4}$
Final Answer : (3)

42) **Question Explanation:** To find the tension at given point of rope.
Concept: Motion of connected bodies.
Formula: Net force = mass × acceleration.
Solution:
a_{sys} = 10 m/s^2
T − 8g = 8a
T = 8 (a + g)
T = 8 (10 + 10)
$\boxed{T = 160N}$
Final Answer : (3)

43) **Explanation:** To find minimum percentage error in given a relation
Concept: This question is based on percentage error
Solution:
$P = \frac{\sqrt{abc^2}}{d^3e^{1/3}}$

$$\frac{\Delta P}{P}\times 100\left[\frac{1}{2}\times\frac{\Delta a}{a}+\frac{1}{2}\times\frac{\Delta b}{b}+\frac{\Delta c}{c}+3\times\frac{\Delta d}{d}+\frac{1}{3}\times\frac{\Delta e}{e}\right]\times 100$$

$$=\left[\frac{1}{2}\times 2\%+\frac{1}{2}\times 3\%+2\%+3\times 1\%+\frac{1}{3}\times 6\%\right]$$

= [1% +1.5% +2% +3% +2%]
The minimum amount of error is contributed by the measurement of a.
Final Answer: (2)

44) **Explaining:** To find value of x and y which are dimension of mass and spring constant in the given relation

Concept: This question is based on dimensional analysis
Solution:
$f = C\ m^x\ k^y$
$T^{-1} = [M]^x\ [MT^{-2}]^y$
$M^0L^0T^{-1} = M^{x+y}T^{-2y}$
Comparing powers of M and T
M: $x + y = 0$
T : $-2y = -1 \Rightarrow y = \frac{1}{2}, x = -\frac{1}{2}$
OR
Time-period of simple pendulum
$T = 2\pi\sqrt{m/k}$
$f = \frac{1}{T} = \frac{1}{2\pi}\left(\frac{k}{m}\right)^{1/2}$
$f = C\ m^x\ k^y$
Comparing powers of m and k
$x = -\frac{1}{2}, y = \frac{1}{2}$
Final Answer: (4)

45) **Explaining:** To match acceleration and velocity at different time as given in the questions
Concept: This question is based on kinematics of motion
Solution:
$V = (t^2 - 2t + 1)$
At $t = 3$ sec, $V = 4$ m/s
$a = \frac{dv}{dt} = 2t - 2$
at $t = 2$ sec, $a = 2$ m/s^2
when $V = 0 \Rightarrow (t^2 - 2t + 1) = 0$
$\Rightarrow t = 1$ sec
$a_{av} = \frac{v-u}{t} = \frac{0-1}{1} = -1$
Final Answer: (3)

CHEMISTRY

46) **Explaining :** Identify the balanced chemical reaction from the given rate expression
Concept : This question is based on rate of reaction in terms of stoichiometry
Solution : Minus signs are for reactants and positive signs for products and dividing numbers are the coefficients.
Final Answer : (2)

47) **Explaining :** Find overall order of the given complex reaction
Concept : This question is based on rate of complex reaction
Solution :
$r = k[A][B_2]$

and $k_c = \frac{[A][A]}{A_2}$ or $[A] = k_c^{1/2}[A_2]^{1/2}$

Thus $r = k \cdot k_c^{1/2}[A_2]^{1/2}[B_2]$

$= k' [A_2]^{1/2}[B]_2$

or order is $\frac{3}{2}$

Final Answer : (3)

48) **Explaining :** Find unit of rate constant from the given rate expression

Concept : This question is based on unit of rate constant

Solution :

Total order $= 1 + \frac{2}{3} = \frac{5}{3}$

so unit of rate constant = $(mol)^{1-\frac{5}{3}}(lit)^{\frac{5}{3}-1}\,sec^{-1}$

Final Answer : (1)

49) **Explaining :** For a first order reaction, if the time taken for completion of 50% of the reaction is t second, the time required for completion of 99.9% of the reaction is :

Concept : This question is based on first order reactions

Solution :

$t = \frac{0.693}{k}$ for first order reaction

$k = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$

$\frac{0.693}{t_{1/2}} = \frac{2.303}{t_{99.9\%}}\log\left(\frac{100}{100-99.9}\right)$

$t_{99.9\%} = \frac{6.903}{0.693} \approx 10\,t$

Final Answer : (2)

50) **Explaining :** For the first order reaction with rate constant k, which expression gives the half-life period (Initial concentration = a) :

Concept : This question is based on half life of first order reaction

Solution : $t_{1/2} = \frac{0.693}{k} = \frac{2.303\log 2}{k} = \frac{\ln 2}{k}$

Final Answer : (1)

51) **Explaining :** Half-lives of first order and zeroth order reactions are same. Ratio of rates at the start of reaction is :

Concept : This question is based on half lives of zero and first order reaction

Solution :

$t_{50}\ (zeroth) = \frac{a}{2k_0}$

$t_{50}\ (first) = \frac{0.693}{k_1}$

$\frac{a}{2k_0} = \frac{0.693}{k_1}$

$$\frac{k_1}{k_0} = \frac{2 \times 0.693}{a}$$

$$\frac{\left(\frac{dx}{dt}\right)_1}{\left(\frac{dx}{dt}\right)_0} = \frac{k_1 a}{k_0}$$

$= 2 \times 0.693$

Final Answer : (2)

52) **Explaining :** The rate constant is numerically the same for three reaction of first, second and third order respectively which one is true for the three of reaction, if concentration of reactant is greater than 1M?

Concept : This question is based on rate law

Solution :

$r_1 = k[A]$ $r_2 = k[A]^2$ $r_3 = k[A]^3$

if $[A] > 1$ then $r_3 > r_2 > r_1$

Final Answer : (3)

53) **Explaining :** An elementary reaction between A and B is a second order reaction. Which of the following rate equations must be correct ?

Concept : This question is based on rate law of an elementary reaction

Solution :

Elementary reaction means a single step reaction

$A + B \rightarrow C$ (experimental reaction)

Final Answer : (4)

54) **Explaining :** If the rate of gaseous reactions is independent of pressure, the order of reaction is :

Concept : This question is based on order of reaction

Solution :

For zero order reaction, r = k (independent of the concentration and pressure of reactants)

Final Answer : (1)

55) **Explaining :** The rate constant of a reaction becomes equal to the pre-exponential factor when :

Concept : This question is based on Arrhenius equation

Solution :

Arrhenius equation

$$\log k = \log A - \frac{E_a}{2.303RT}$$

when $T \rightarrow \infty$ then $\log k = \log A$

or $k = A$

Final Answer : (1)

56) **Explaining :** Rate constant depends on

Concept : This question is based on rate constant

Solution : Rate constant depends on presence of catalyst and temperature

Final Answer : (1)

57) **Explaining :** Match the column-I with Column-II :
Concept : This question is based on prediction of order from graph
Solution :
Zero order ; $[A]_t = [A]_0 - kt$
First order ; $r = k[A]$
Second order ; $\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$
Third order ; $r = k[A]^3$
Final Answer : (2)

58) **Explaining :** Match the half-lives in Column-I with the orders in Column-II :
Concept : This question is based on half life
Solution :
Zero order ; $t_{1/2} = \frac{a_0}{2k}$
First order ; $t_{1/2} = \frac{0.693}{k}$
Second order ; $t_{1/2} = \frac{1}{a_0 k}$
Final Answer : (1)

59) **Explaining :** Calculate the rate constant of given reaction
Concept : This question is based on calculation of rate constant
Solution :
Ratio of equation $\frac{(1)}{(2)}$ $\quad \frac{1}{2} = \left[\frac{1}{2}\right]^m$
$\therefore$ m = 1
Ratio of equation $\frac{(2)}{(3)}$ $\quad \frac{1}{2} = \left(\frac{1}{2}\right)^n$
n = 1
So rate equation $r = k[A_2]^1 [B_2]^1$
Order = 1 + 1 = 2
Final Answer : (1)

60) **Explaining :** The process of reaction can be described by which of the following graphs?
Concept : This question is based on graph of exothermic reactions
Solution :
For an exothermic reaction, the potential energy of the products (AB) must be lower than the potential energy of the reactants (A + B).
Since, the reaction involves an intermediate (X), the graph must have two peaks. The first step, being fast needs less activation energy. The second step being slow, needs more activation energy.
$\therefore$ second peak will be higher than the first peak.
The correct graph is (3)

Final Answer : (3)

61) **Explaining :** Match the Column-I with Column-II :
Concept : This question is based on Van't hoff factor
Solution :
For dissociation, i > 1
(A) $Ca_3(PO_4)_2 \rightarrow 3Ca^{2+} + 2PO_4^{3-}$; i = 1 + (n-1) α
assuming 100% dissociation i = n = 5
(B) $Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$
i = 5
(C) Urea → i = 1
(D) CH_3COOH in benzene associates to form dimer i < 1
Final Answer : (2)

62) **Explaining :** Freezing point of an aqueous solution is –0.186°C.
Concept : This question is based on elevation in boiling point
Solution :

$$\Delta T_f = \frac{1000 \cdot K_f \cdot w}{W \cdot m}$$

and $\Delta T_b = \frac{1000 \cdot K_b \cdot w}{W \cdot m}$

$$\therefore \frac{\Delta T_f}{\Delta T_b} = \frac{K_f}{K_b} \text{ or } \frac{0.186}{\Delta T_b} = \frac{1.86}{0.512}$$

$\therefore \Delta T_b = 0.0512°C$
Final Answer : (2)

63) **Explaining :** The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is:
Concept : This question is based on Van't Hoff factor for dissociation
Solution : For $Ba(OH)_2$, i = n = 3
Final Answer : (4)

64) **Explaining :** A 5% (w/v) solution of cane sugar (molecular weight = 342) is isotonic with 1% (w/v) solution of a non-electrolyte substance X. The molecular weight of X is:
Concept : This question is based on isotonic solutions
Solution :
For Isotonic $C_1 = C_2$ (at same temperature)

$$\frac{w_1}{M_{w_1} \times V_1} = \frac{w_2}{M_{w_2} \times V_2}$$

$$\frac{5}{342 \times 100} = \frac{1}{M_{w_2} \times 100}$$

$$M_{w_2} = 68.4$$

Final Answer : (2)

65) **Explaining :** 0.5 M aqueous solution of glucose is isotonic with:
Concept : This question is based on isotonic solutions
Solution : For isotonic solutions ; $\pi_1 = \pi_2$ or $ic_1 = ic_2$ (at same temperature)

Final Answer : (3)

66) **Explaining :** Which of the following is dependent on temperature?
Concept : This question is based on concentration terms
Solution : Molarity is volume-dependent so it changes with temperature
Final Answer : (1)

67) **Explaining :** Normality of 10% (w/v) H_2SO_4 solution nearly :
Concept : This question is based on normality
Solution :

$$N = \frac{w \times 1000}{E_A \times V_{mL}} = \frac{10 \times 1000}{49 \times 100}$$

$N \simeq 2$
Final Answer : (4)

68) **Explaining :** 5 N H_2SO_4 was diluted from 1 litre to 10 litres. Normality of the solution obtained is:
Concept : This question is based on concept of dilution
Solution :
$N_1V_1 = N_2V_2$
$5 \times 1 = N_2 \times 10$
$N_2 = 0.5$ N
Final Answer : (4)

69) **Explaining :** OR will be equal to :
Concept : This question is based on Raoult's Law
Solution : $P_T = P_A + P_B$ or OR = OQ + OP
Final Answer : (3)

70) **Explaining :** Which one is not equal to zero for an ideal solution:
Concept : This question is based on ideal solutions.
Solution : $\Delta S_{mix} > 0$
Final Answer : (1)

71) **Explaining :** For an ideal solution of two components A and B, which of the following is true?
Concept : This question is based on ideal solutions
Solution :
For an ideal solution:
$\Delta H_{mix} = 0$; $\Delta S_{mix} > 0$
$\Delta V_{mix} = 0$; $\Delta G_{mix} < 0$
A-A interactions $\simeq$ B-B interactions $\simeq$ A-B interactions
Final Answer : (3)

72) **Explaining :** The rate constant of most of the reactions increases with increase in

temperature. According to the Arrhenius equation, $k = Ae^{-E_a/RT}$. The curve of rate constant 'k' against temperature 'T' will be :
Concept : This question is based on Arrhenius equation
Solution : $k = Ae^{-E_a/RT}$
Final Answer : (4)

73) **Explaining :** The rate constant of a reaction is 2.1×10^{-2} mol^{-2} $litre^2$ min^{-1}. The order of reaction is:
Concept : This question is based on unit of rate constant
Solution :
Unit of rate constant
$k = mol^{1-(m+n)}$ $lit^{(m+n-1)}$ $time^{-1}$
Final Answer : (3)

74) **Explaining :** Find the rate law expressions
Concept : This question is based on rate of complex reactions
Solution :
From slow step
$r = k [O] [O_3]$
[O] is intermediate so it will be replaced by using another step (fast step)

$$K_c = \frac{[O_2][O]}{[O_3]} \quad \text{or} \quad [O] = K_c\frac{[O_3]}{[O_2]}$$

then $$r = kK_c\frac{[O_3]^2}{[O_2]}$$
Final Answer : (4)

75) **Explaining :** The temperature coefficient for most of the reactions lies between:
Concept : This question is based on temperature coefficient
Solution : Temperature coefficient has a value between 2 and 3 for most reactions
Final Answer : (2)

76) **Explaining :** The reaction $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ is of first order. If volume of reaction vessel is reduced to $\frac{1}{3}$, the rate of reaction would be :
Concept : This question is based on rate law
Solution :
$\text{Conc}^n. \propto \frac{1}{\text{Volume}} \Rightarrow C' = 3C$
$r \propto C$ ...(i)
$r' \propto 3C$ ...(ii)
Divide i and ii, we get $r' = 3r$
Final Answer : (3)

77) **Explaining :** If in a 1st order reaction 75% of reaction complete in 4 hrs. Then how much time is required to complete 87.5% of reaction?
Concept : This question is based on fractional lives of first order reaction.
Solution :

For 75% $t_{75\%} = 2 \times t_{1/2}$ ...(i)
For 87.5% $t_{87.5\%} = 3 \times t_{1/2}$ ...(ii)
on there ratio $\frac{4}{t_{87.5}} = \frac{2}{3}$

$$t_{87.5} = \frac{4 \times 3}{2} = 6\ hr$$

Final Answer : (2)

78) **Explaining :** When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is :
Concept : This question is based on determination of order of reaction
Solution : For first order reaction ; $t_{1/2} = \frac{0.693}{k} \quad (t_{1/2} \propto a^0)$
Final Answer : (1)

79) **Explaining :** Slope of the straight line obtained by plotting $\log_{10}$ k against 1/T represents which term?
Concept : This question is based on Arrhenius equation
Solution :
$k = Ae^{-E_a/RT}$
or $\log k = \log A - \frac{E_a}{2.303RT}$
$y = c + m x$
$\therefore$ slope $= = \frac{-E_a}{2.303R}$
Final Answer : (3)

80) **Explaining :** If 18 g of glucose is present in 1000 g of solvent, the solution is said to be:
Concept : This question is based on molality
Solution : $m = \frac{w_A \times 1000}{m_A \times w_B} = \frac{18 \times 1000}{180 \times 1000} = 0.1\ molal$
Final Answer : (4)

81) **Explaining :** The depression in f.pt. of 0.01 m aqueous solution of urea, sodium chloride and sodium sulphate is in the ratio:
Concept : This question is based on depression in freezing point
Solution :
Net Molarity or Molality = m × no. of particles
Urea : NaCl : Na_2SO_4
0.01×1 0.01×2 0.01×3
1 : 2 : 3
Final Answer : (3)

82) **Explaining :** Vapour pressure of a solution containing non-volatile solute is:
Concept : This question is based on vapour pressure
Solution :
V.P. of solution = V.P. of solvent in the solution

which is directly proportional to mole fraction of solvent in the solution.
Final Answer : (2)

83) **Explaining :** Value of Henry's constant (K_H)__.
Concept : This question is based on Henry's law
Solution : Value of Henry's constant depends on nature of gas and temperature. K_H increases with increase in temperature.
Final Answer : (2)

84) **Explaining :** If molality of the dilute solutions is doubled, the value of molal depression constant (K_f) will be:
Concept : This question is based on molal depression constant
Solution : k_f depends only on nature of solvent.
Final Answer : (3)

85) **Explaining :** The mixture of two liquids boils without changing its composition is called:
Concept : This question is based on Azeotropic mixtures
Solution :
Azeotropes are binary liquid mixtures having same composition in liquid and vapour phase and boil at a constant temperature
Final Answer : (3)

86) **Explaining :** What is the freezing point of a solution that contains 10.0 g of a glucose ($C_6H_{12}O_6$) in 100 g of H_2O ?
Concept : This question is based on depression in freezing point
Solution :

$$\Delta T_f = \frac{K_f \times W_A \times 1000}{M_A \times W_B} = \frac{1.86 \times 10 \times 1000}{180 \times 100}$$

$T_f - T' = 1.03$
$0 - T' = 1.03$
$T' = -1.03$
Final Answer : (4)

87) **Explaining :** Which of the following is not a non-electrolyte?
Concept : This question is based on electrolytes
Solution : Acetic acid is a weak electrolyte, so it is not a non-electrolyte.
Final Answer : (1)

88) **Explaining :** Pure water can be obtained from sea water by:
Concept : This question is based on reverse osmosis
Solution : Reverse osmosis is used in desalination of sea water.
Final Answer : (3)

89) **Explaining :** What is the van't Hoff factor for sodium phosphate?
Concept : This question is based on Van't Hoff factor
Solution : Na_3PO_4 i = n = no. of ions = 4

Final Answer : (4)

90) **Explaining :** Van't Hoff factor more than unity indicates that the solute in solution has:
Concept : This question is based on Van't Hoff factor
Solution : In case of dissociation, value of Van't Hoff factor (i) is greater than unity.
Final Answer : (2)

BIOLOGY

91) **Explanation:** A–ii, B–iii, C–i, D–iv

A. Squamous epithelium	ii. Alveoli, blood vessels – diffusion
B. Cuboidal epithelium	iii. Kidney tubules, ducts – secretion & absorption
C. Columnar epithelium	i. Intestine, stomach – secretion & absorption
D. Ciliated epithelium	iv. Bronchioles, fallopian tube – movement of mucus/ovum

Final Answer : (1)

92) **Explanation:** Secretory vesicles first formed → secretion occurs → ducts carry products → released to specific site.
Final Answer : (1)

93) **Explanation:**
2 Incorrect → Stores fats (triglycerides), not proteins.
5 Incorrect → Fibroblasts secrete fibres & ground substance.
Final Answer : (3)

94) **Explanation:**
Cartilage = pliable, resists compression, chondrocytes in cavities.
Bone = hard, calcium salts + collagen, structural support.
Blood = fluid tissue, transport medium.
Final Answer : (4)

95) **Explanation:**
Skeletal muscle (a) → striated, multinucleated, voluntary.
Cardiac muscle (c) → striated, intercalated discs (junctions).
Smooth muscle (b) → fusiform, non-striated, involuntary.
Axon (d) → carries impulses away from the cell body.
Final Answer : (1)

96) **Explanation:**
Statement 1 False – it may be unicellular (e.g., goblet cell).
Statement 5 False – compound epithelium is multi-layered, protective not absorptive.
Final Answer : (2)

97) **Explanation:** Both A and R are correct but reason is not the correct explanation of A.
Final Answer : (2)

98) **Solution :** Ligaments have dense regular connective tissue
Final Answer : (1)

99) **Explanation:**
Statement (2) is false → Adipose tissue (b) specialised to store fats in fat storage areas.
Statement (3) is false → plasma membrane is present in all cells.
Final Answer : (1)

100) **Explanation:** 1–B, 2–C, 3–D, 4–A
Filiform antennae = sensory organ.
Compound eyes = vision with ommatidia.
Mesothoracic leg = locomotion.
Tegmina = first pair of forewings, leathery, protective.
Final Answer : (4)

101) **Explanation:** (3) Incorrect → Labrum = upper lip, Labium = lower lip.
Final Answer : (2)

102) **Explanation:**
(2) Incorrect → Malpighian tubules are excretory, not digestive.
(3) Incorrect → Cockroach has an open circulatory system.
Final Answer : (3)

103) **Explanation:**
Testis (T): Produces sperms.
Vas deferens (B): Transports sperms from testis.
Seminal vesicle (A): Stores sperms.
Ejaculatory duct (E): Ejects sperms out during copulation.
Final Answer : (1)

104) **Explanation:** Statement (2) False – antennae arise in front of the eyes, not behind.
Final Answer : (4)

105) **Explanation:** Mouth → Pharynx → Oesophagus → Crop → Gizzard → Midgut → Ileum → Colon → Rectum → Anus
Final Answer : (2)

106) **Explanation:**
(i) The liver secretes **bile** which is stored in the **gall bladder**.
(ii) The **pancreas** produces pancreatic juice containing digestive enzymes.
(iii) Undigested food passes into the **rectum** and finally opens outside by the cloaca.

Final Answer : (1)

107) **Explanation:** Statement 1 Incorrect → During aestivation/hibernation, respiration is mainly cutaneous (through skin), not lungs. Statement 3 is incorrect → Hepatic portal system connects the intestine and liver.
Final Answer : (4)

108) **Explanation:** A–ii, B–iii, C–i, D–iv
Forebrain: olfactory lobes, cerebral hemispheres → voluntary actions, smell, intelligence.
Midbrain: Pair of optic nerves.
Hindbrain (cerebellum) → balance & equilibrium.
Medulla oblongata → involuntary functions like heartbeat & breathing.
Final Answer : (1)

109) **Explanation:**
(3) Incorrect → Fertilization is external in water.
(5) Incorrect → Tympanum helps in hearing + balance, but external ear is absent.
Final Answer : (1)

110) **Explanation:** Hind limbs are actually used for swimming, leaping, and locomotion, not mainly for burrowing.
Final Answer : (3)

111) **Explanation :** Tight junctions seal adjacent cells together to prevent leakage.
Adhering junctions perform cementing, and Gap junctions facilitate communication.
Final Answer : (2)

112) **Explanation :** Cuboidal epithelium is found in duct of glands and tubular parts of nephrons in kidney.
Final Answer : (3)

113) **Explanation :** The inner lining of blood vessels is called the endothelium, which is made of a single layer of flat squamous epithelial cells.
Final Answer : (3)

114) **Solution :**

A. a. Anal cerci → (iii) 10th tergum

A. b. Anal style → (iv) 9th Sternum

A. c. Testis → (i) 4-6th abdominal segment

A. d. Ovary → (v) 2-6th abdominal segment

A. e. Mushroom gland → (ii) 6-7th abdominal segment

Final Answer : (2)

115) **Solution :** Statement IV is incorrect.
Segmental pores present in thorax and abdomen are called spiracles, not sclerites, that help in respiration.
Final Answer : (3)

116) **Solution:** The hard, chitinous plates that make up the exoskeleton of a cockroach are called sclerites. These plates are present in all body segments and are joined by a flexible, membranous cuticle called the arthrodial membrane.
Final Answer : (2)

117) **Solution:** The hypopharynx is a median, flexible, tongue-like lobe located between the mandibles in the mouth of a cockroach.
Final Answer : (3)

118) **Solution:** Metamorphosis is the biological process of a tadpole transforming into an adult frog. This includes significant physical changes like the development of limbs, the disappearance of the tail, and changes in the respiratory system.
Final Answer : (1)

119) **Explanation :** Cutaneous respiration is the exchange of gases that takes place through the moist, permeable skin of the frog, both in water and on land.
Pulmonary respiration is the exchange of gases through the lungs, which is the primary mode of breathing on land.
Final Answer : (1)

120) **Solution:** The frog has both a hepatic portal system (veins from the digestive tract going to the liver) and a renal portal system (veins from the hindlimbs and pelvic region going to the kidneys), which is a characteristic feature of amphibians.
Final Answer : (1)

121) **Solution :** A – Epidermis : Outermost layer of microsporangium.
B – Endothecium : Inner to epidermis
C – Middle layers :
D – Tapetum : : Innermost layer → nourishes developing pollen grains
Final Answer : (2)

122)

Explain Question : The question is asking about: Identifying the labeled parts A, B, and C in the diagram of an ovule.

Concept : Ovule Structure.

Solution : The diagram shows the ovule attached to its stalk:

A. A points to the fusion point between the funicle and ovule body, the Hilum.

B. B points to the stalk attaching the ovule to the placenta, the Funicle.

C. C points to the mass of nutritive cells, the Nucellus.

Final Answer : Option (3) A-Hilum, B-Funicle, C-Nucellus

123) **Solution :** B is cleistogamous flowers which prevents xenogamy but not autogamy
Final Answer : (3)

124) **Solution :** Both true
Anther = bilobed, tetrasporangiate
T.S. = circular
Final Answer : (4)

125) **Solution :** Both **Statement I** and **Statement II** are correct according to NCERT
Final Answer : (4)

126) **Solution :** Statement I correct
Autogamy **requires synchrony** in pollen release and stigma receptivity.
Final Answer : (2)

127) **Solution :** Both correct
Central cell contains polar nuclei.
Only one megaspore (n) remains functional one of any of is develops into single embryo sac.
Final Answer : (1)

128) **Question Explanation:** Pollen viability in cereals.
Concept: Pollen Viability
Solution: Pollen viability is the ability of pollen grains to germinate and fertilize the ovule. In cereals like **wheat** and **rice**, the pollen grains have a very short period of viability, losing their ability to function after approximately **30 minutes** of being released from the anther.
Final Answer: 30 minutes

129) **Solution :**

A. (i) **False** **:** Main plant body = **sporophyte**, not gametophyte
B. (ii) **True** **:** Thalamus = modified stem
C. (iii) **True** **:** Nodes/internodes absent → correct but tricky (not counted here)
D. (iv) **False** **:** Angiosperm endosperm = Triploid

Final Answer : (3)

130) **Question Explanation :** The question asks about the cellular origin of male gametes (sperm cells) in flowering plants.

Concept :
This question is based on structure of the male gametophyte (pollen grain).
Explanation:
"In a mature pollen grain, the generative cell divides mitotically to give rise to two male gametes."

Final answer : Generative cell

131)

Question Explanation: Question is asking about the what is the common structure or type of female gametophyte found in most angiosperms

Concept: Female gametophyte

Solution:

In a mature embryo sac three cells are grouped together at the micropylar end and constitute the egg apparatus. The egg apparatus, in turn, consists of two synergids and one egg cell. Three cells are at the chalazal end and are called the antipodals. The large central cell has two polar nuclei. Thus, a typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled

Final answer: 2

132)

The correct answer is: **A is true but R is false.**
Solution :
Assertion: True. In cereals like rice and wheat, pollen grains lose viability within 30 minutes of their release.
Reason: False. Pollen grains of families like Rosaceae, Leguminosae, and Solanaceae can maintain viability for days to months, not just 30 minutes.
Final Answer: (3)

133) **Solution :** Pollen tablets and syrups are used by athletes and race horses.
Final Answer : (1)

134)

Question Explanation : Identifying the plant pair whose structural mechanism prevents both autogamy and geitonogamy.

Concept : This question is based on outbreeding devices.

Solution: The condition of dioecy (male and female flowers on separate plants) physically eliminates all forms of self-pollination. Papaya and Datepalm are dioecious plants.

Final Answer : option (4) Datepalm & Papaya

135) **Solution :** Needed when **female parent is bisexual**
 Remove anthers to prevent self-pollination
Final Answer : (1)

136) **Solution :** Colourful flowers
 Foul odour (in some cases)
 Large flowers

Incorrect options:

A. Non-sticky pollen
B. Well-exposed stamens (wind feature)
C. Mucilage (water pollination)

Final Answer : (1)

137) **Solution : Cleistogamous flowers** → always self-pollinate → assured seed set
Final Answer : (3)

138) **Solution : Feathery stigma + exposed stamens + many flowers** → wind pollination traits
Final Answer : (2)

139) **Solution :**

A. Nectar & pollen = rewards
B. Reason is true.

Final Answer : (3)

140) **Solution :** Archesporium → megaspore mother cell → megaspore → embryosac
Final Answer : (4)

141) **Solution :**

A. Hydrophily → **Zostera**
B. Anemophily → **Grasses**
C. Zoophily → **Amorphophallus**

Final Answer : (3)

142) **Solution :** The two cells at which pollen grains are discharged are **s**maller generative cell and larger vegetative cell.

Final Answer : (4)

143) **Solution :** □ Dichogamy (timing difference)
□ Herkogamy
□ Self-incompatibility
Final Answer : (3)

144) **Solution :** Statement A incorrect → Vallisneria is **water pollinated**, not wind
Final Answer : (1)

145) **Solution :**

A. Cleistogamy → bud pollination
B. Geitonogamy → same plant
C. Entomophily → insects
D. Xenogamy → cross-pollination

Final Answer : (3)

146) **Explanation:**
– Amorphophallus (P) is insect-pollinated: A-P
– Maize (Q) is wind-pollinated: B-Q
– Vallisneria (R) is pollinated on water surface: C-R
– Hydrophily (Hypohydrophily) -Zostera
Concept: Pollination agents and examples
Final Answer : (1)

147) **Solution :** Antipodal cell, Male gamete, Pollen grain, Synergids, Megaspore, microspore, tetrad
Final Answer : (2)

148) **Solution :** Flowers have **both essential & non-essential whorls** → statement incorrect
Final Answer : (3)

149) **Solution :** Insects
Final Answer : (3)

150) **Solution :**

A. Nucellus = 2n
B. MMC = 2n
C. Functional megaspore = n
D. Female gametophyte = n

Final Answer : (3)

151) **Solution :** Unicellulars are capable of independent existence. Leeuwenhoek discovered

living cell in 1674.
Final Answer : (2)

152) **Solution :** Prokaryotic cells lack centrosomes.
Final Answer : (3)

153) **Solution :** Ribosomes are present in both prokaryotic & eukaryotic cells.
Final Answer : (1)

154) **Solution :** A-3, B-4, C-1, D-2

Column-I	Column-II
A. Mycoplasma	3. 0.3 μm in length
B. Human RBCs	4. About 7.0 μm in diameter
C. Bacteria	1. 3 to 5 μm in length
D. Eukaryotic cell	2. 10 to 20 μm in size

Final Answer : (4)

155) **Solution :** Both A and R are true, and R is the correct explanation of A.
Final Answer : (1)

156) **Solution :** Cell wall + plastids + large central vacuole
Final Answer : (2)

157) **Solution :** All are correct.
Final Answer : (4)

158) **Solution :** Both A and R are true, and R is the correct explanation of A.
Final Answer : (1)

159) **Solution :** Cell membrane property of fluidity plays important role in cell secretion & cell division.
Final Answer : (3)

160) **Solution :** A polar molecule needs to cross the plasma membrane. Direct passage through the nonpolar lipid bilayer is not possible for this molecule. It would more through carrier protein or channel protein mediated transport.
Final Answer : (2)

161) **Solution :** Mitochondrion is not included in the endomembrane system.
Final Answer : (4)

162) **Solution :** Both A and R are true, and R is the correct explanation of A.
Final Answer : (1)

163) **Solution :** Vacuole — tonoplast
Final Answer : (2)

164) **Solution :** All statements are correct.
Final Answer : (4)

165) **Solution :** Axoneme is covered by plasma membrane.
Final Answer : (3)

166) **Solution :** All statement are correct.
Final Answer : (4)

167) **Solution :** All statement are correct.
Final Answer : (4)

168) **Solution :** Metacentric → Acrocentric → Sub-metacentric → Telocentric
Final Answer : (3)

169) **Solution :** RNA and protein can move in both direction.
Final Answer : (2)

170) **Solution :** Nucleolus is non-membrane bound structure.
Final Answer : (2)

171) **Solution :** Ribosomal subunits are non-membrane bound.
Final Answer : (4)

172) **Solution :** Lysosome possess hydrolytic enzymes.
Final Answer : (4)

173) **Solution :** Both A and R are true, but R is the correct explanation of A.
Final Answer : (1)

174) **Solution :** A-R, B-P, C-Q, D-S

Column I	Column II
A. Glycocalyx	R. May occur as slime layer or capsule
B. Cell wall	P. Determines bacterial shape and provides structural support

C. Plasma membrane	Q. Selectively permeable and interacts with the outside world
D. Plasmid	S. Small circular DNA outside genomic DNA

Final Answer : (3)

175) **Solution :** Active transport requires energy.
Final Answer : (4)

176) **Solution :** Loose outer sheath – Slime layer, thick & tough outer covering – Capsule
Final Answer : (2)

177) **Solution :** Non-polar molecules pass through non-polar lipid bilayer.
Final Answer : (4)

178) **Solution :** Fimbriae – help bacteria in attachment
Final Answer : (3)

179) **Solution :** Glyoxysome is a membrane-bound minute vesicle belonging to the group of microbodies.
Final Answer : (1)

180) **Solution :** Plasmid, 70S ribosome, pili/fimbriae
Final Answer : (2)